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Table of Contents

Executive Summary **4**

1. The Dependency Issue **7**
1.1 Why Europe Needs a Non-Dependence Strategy 7
1.2 Dependency Situations 10
1.3 Levels of Dependence Criticality 12
1.4 General Context 14
1.4.1 Global Supply Chain 14
1.4.2 Space Programmes in Asia 16
1.5 Technology and Innovation Policy 16
1.5.1 Overall View on Space Technology, the Innovation Process and the European Actors 16
1.5.2 ESA Technology Policy – Generic Programmes 20
1.6 Industrial Policy and Competitiveness 24
1.7 Key Enabling Technologies: New Opportunities for Space 25
1.8 Defence and Security 26

2. The Long Road toward a European Technological Non-Dependence Strategy **29**
2.1 Non-Dependence Initiatives within ESA 29
2.1.1 ESA Technological Harmonisation Programme 29
2.1.2 European Space Component Coordination - ESCC 32
2.1.3 European Component Initiative – ECI 38
2.2 Non-Dependence Initiatives within European Union 39
2.2.1 European Space Technology Platform - ESTP 39
2.2.2 The 7th Framework Program: First Steps 41
2.2.3 Horizon 2020: The New Framework Program for Research and Innovation 42
2.3 EC-ESA-EDA Joint Task Force for European Technological Non-Dependence (ETnD) 43

3. Critical Technologies: State of Play **46**
3.1 Successful Examples of Harmonised Critical Technologies 46
3.2 Current Technological Priorities 46
3.3 Status and Evolution of Non-Dependence Situation 47
3.3.1 Technological Domains and the Dependency Issue 48
3.4 Main Conclusions of the ESCCON Conference 2013 50

4. Conclusions and Recommendations **51**

Annex **53**
A.1 European Component Initiative (ECI): The Four phases 53

Acknowledgements **55**

About the Author **55**

Executive Summary

This report is an overview of European technological Non-Dependence in the space field. The general context, what has been already done and what is still to be exploited and pursued, is presented.

After an introduction to the Non-Dependence issue, the first chapter defines the overall context of the issue different perspectives. In this part the most relevant cases of dependence are described, as are the international scenario and the new dynamics in space; the innovation process, with the main critical issues in Europe, and finally the security issues.

In the second chapter all the European initiatives taken toward technological Non-Dependence are described. Thus the first section sets out the initiatives managed by ESA (the Harmonisation programme, the European Space Component Coordination (ESCC) and the European Component Initiative (ECI)); in the second section the programmes of European Union are presented (the European Space Technology Platform (ESTP), and some specific actions under the Seventh Framework Programme for Research (FP7) and within Horizon 2020). The third section contains a description and discussion of the Joint Task Force (JTF) for Technological Non-Dependence established between EC, ESA and EDA in 2008.

In the third chapter a detailed description of the current critical technologies is given. In the first section, some successful examples of harmonised technologies are provided, and in the second section the macro-areas relevant for future development and funding are described. Finally, in the third section, specific technologies in which Europe is strongly dependent are presented with a detailed analysis.

At the end conclusions and recommendations are given as follows:

- Eropean Technological Non-Dependence is an overarching topic that must be considered under many different perspectives. Technological dependence hampers space utilisation, security and defence, productivity and market competitiveness.
- Even though some technology gaps between Europe, the U.S. and Asia have been reduced, the gap is still much too big for Europe to be independent. This imposes a challenge for foreign policy and international cooperation.
- The enhancement of a European Supply Chain through technological development should be pursued, but it should proceed in parallel with international relations and policy actions (e.g. the example of fruitful cooperation with Japan).
- Many initiatives and actions have been taken both within ESA and at institutional level, finally converging in the Joint Task Force of the EC, ESA and EDA for European Technological Non-Dependence. Nevertheless, an overall programmatic strategy at European level is still lacking.
- The first driver for European space technology policy should be security of supply. It should be pursued by two approaches: on the one hand, the enhancement of the European supply chain; on the other side, the intensification of international collaboration and exchanges. The management of the supply chain should be a primary focus in the management of space programmes.
- The establishment of the Joint Task Force for Non-Dependence has as its goal improvement in the availability of components and building blocks. This is necessary because of the weakness of European industry in this field and the strategic importance of these lower levels of the supply chain. The Urgent Actions defined by the Joint Task Force promoted in this direction are an important first step.
- At a technological level, the first priority is to manage medium/long term obsolescence. It is necessary to achieve the right balance between innovation and product maintenance. To do this, innovation timeframes have to become more compatible with market dynamics.
- European technological strategies focus principally on product concepts and the early stages of innovation. More attention must be given to maturity and readiness aspects (high TRL). This is the guarantee that the European space industrial base is able to supply the tech-

nologies required, and thus a crucial step in remaining competitive in the global market. In this regard, qualification and in-orbit validation are fundamental steps. Strong efforts and investment should be applied to these later phases of the technological process, to bring innovative technologies to the right level for risk-free implementation in European programmes and for commercialisation. Intensifying the frequency of control and qualification activities will bring more efficiency in the technological process and in the planning capability.

- The periodic preparation of detailed technical roadmaps, marking the evolution of activities and perspectives, is a very important tool, especially when considering the fragmented European reality and the large number of stakeholders involved. Similarly, it is essential to standardise technologies and procedures. The definition of standards and of precise final requirements is a precondition for transnational and international collaboration, both within EU and with third countries.
- A high level of reliability is one of the main characteristics of European technologies and missions, and actually contributes to giving Europe an outstanding position in the global space arena. This is an achievement that should be maintained and improved, although it requires long timeframes and expensive technical processes.
- The current European fragmentation, at national and industrial level, strongly affects the technological scenario. More coordination is needed, with tailored actions, joint undertakings, strong synergistic efforts and technology transfer plans. Technology planning and coordination is fundamental, as is the elaboration of long-term prospects for breakthrough concepts and associated technologies. The involvement of all the different actors in a structured multi-stakeholder approach, especially in the current situation, is crucial for achieving simplification, and through this effectiveness and progress.
- Europe should invest in its human capital to ensure not only availability, but also (and mostly) capability. This is the key to maintaining a central position in the global context, also in the light of the huge investments in human capital of the emerging Asian space powers.
- To boost the development of the space sector, stakeholders must make use of all EU, national and regional initiatives that promote synergy in space activities. Since space assets are integrated systems of many different components, a cross-sectorial approach is essential. Spin-in activities should be promoted and enhanced, in a fully open innovation approach. This is relevant to increase the achievement of Non-Dependence goals. Within the ambit of ESA there are many programmes devoted to such cross-cutting promotion. It is necessary to continue to invest in this direction. A well-defined planning of such synergistic actions should be pursued. As concerns components, nanotechnologies and advanced materials, space activities can leverage the many actions proposed by the EC on *Key Enabling Technologies* (KETs), for example with joint calls. In this respect, the role and responsibilities of the ESTP are still insufficiently exploited.
- Public support for R&D in space technologies and long-term space programmes is essential because of the high costs, high level of risk and the insufficient returns from commercial markets. All the different funding sources available should be streamlined and harmonised to achieve more appropriate and effective mechanisms, tailored for longer timeframes and specific programmes.
- Regional and national funding should be harmonised with the technology plans of national space agencies.
- In addition to security of supply, and thus the availability of technology, the sustainability of product lines must be ensured. A sustainable market is a fundamental condition for being competitive. Domestic European markets are limited and fragmented. They need to be aggregated. This should include not only the national space markets, but also in combination with other collateral sectors, first of all defence. Synergies between the civil and defence community are still not being sufficiently exploited. They should be promoted and improved to attain a fully effective Non-Dependence strategy.
- European industrial policy strongly affects technological Non-Dependence. The inefficient translation of the knowledge base into products and services, the excessively long time to market, as well as the fragmentation of industrial and regulatory national systems are among the most important factors that directly lead to obsolescence and dependency issues.

The combined product policy to link technology evolution to user requirements and to procurement programmes must be enhanced.

- *Horizon 2020*, the framework programme for research and innovation, aims to promote the *triangle of knowledge* by also looking at the commercialisation phase, including through better involvement of SMEs. In the establishment of a Non-Dependence strategy at European level, all the new possibilities provided by the new programme should be fully utilised.
- Although the most innovative and advanced items are generally produced by a small number of supplying countries and companies, in recent years the availability of many technologies has been growing. This has modified the approach to export controls in Europe, also imposing the participation in multilateral export controls regimes. The European export control regime is unique, because it combines EU-wide principles with national regulations. However, harmonisation is still not satisfactory. In the Member States there are different types of export licensing and different terms and conditions; this is the result of policy variations and regulatory differences (beside different technological advancement). This situation brings licensing delays for the EU that undermine the fulfilment of the requirements of the export industry and international commitments, hindering market and industrial development. Harmonisation of national legal frameworks should be combined with flexibility and consistency at European level, in order to achieve the best balance between security and productivity.
- The establishment of a well-balanced and harmonised regulatory framework is at the basis of the evolution of any Non-Dependence strategy. This refers firstly to export controls, but procurement regulations and intellectual property rights are also crucial in this specific context.

1. The Dependency Issue

Non-Dependence is an overarching topic, and the situations contributing to dependence are very different, giving rise to a complex scenario. In this chapter a general overview is presented. First Non-Dependence is defined and broadly described, with its different causes (second section) and levels of criticality (third section). The difference between *Non-Dependence* and *Independence* is stressed, and the reasons why Independence is unachievable are described. The following section sets out the general context and the main features of the global supply chain for space are discussed. Due to the increasing importance of Asian powers in the global technology domain special attention has been given to the evolution in this region and a separate section included. Thereafter an overview of space technology and the innovation process is provided, followed by a brief summary of ESA programs for research and technology development. The following section is devoted to industrial policy and its consequences for Non-Dependence. This section also include a description of KETs (Key Enabling Technologies). Finally, the last section is devoted to defence and security, and the many implications for a Non-Dependence strategy of the inherently sensitive character of space technologies.

1.1 Why Europe Needs a Non-Dependence Strategy

While Europe possesses the broad majority of the technologies, processes and industrial capabilities needed to develop space programmes, European stakeholders nonetheless still need to externally source certain components, raw and advanced materials as well as some basic technologies and building blocks that are not available within European boundaries.[1] Although one third of all new space patents are filed in Europe[2], it has been estimated that Europe is five years behind in a number of technologies (for example in microelectronics[3]). At present, the European space industry has good mastery of most of the supply chain, except at the lowest levels with basic supplies, in particular *Electrical, Electronic and Electromechanical* (EEE) components and advanced materials. This clearly places Europe in a position of strong dependence, since the lower the level of technology on which a country or region is dependent, the greater the weakness of the country or region. In fact, this condition of dependency exposes Europe to many different risks and potential uncertainties of supply. Thus Europe is vulnerable to political instability in a supplier country and to natural disasters in any part of the world, because of the global dimension typical of any technological supply chain in current times. Security of supply is not only threatened by political crisis and natural calamity, but also by the regulatory barriers of export controls. Because of the dual-use and highly sensitive nature of most space technologies, they are often subject to export restrictions as strategic and defense-related items. This makes some of them completely unavailable from exporters, mostly the U.S., or creates lengthy procurement delays that can have serious effects on space programmes, especially in the context of the global market.

Export restrictions may apply to basic technologies, products, materials and processes.[4] Referring to technologies under the *International Traffic Regulation of Arms* (ITAR), a set of government regulations that controls the export and import of defence-related items on the *United States Munition List* (*USML*) an industry survey by ESA on commercial space programmes identifies 111 items under restrictions, most of which had issues with a strong impact on their procurement. The analysis showed that for 43% of all restricted items industry saw a strong likelihood of increased delays and costs. Technology was blocked for some users (for 38% of the items) and quality and access to technical information (for 35% of the items) were also identified as important issues. Moreover,

[1] *European Space Technology Master Plan* 2011, hereinafter ESTMP 2011
[2] *The Space Economy at a glance 2011*, OECD, Directorate of Science, Technology and Industry, http://www.oecd.org/sti/futures/space/48301203.pdf
[3] *FP7-Space: R&D in support of European microelectronics enabling technology*, Richard Gilmore, ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011
[4] ESTMP 2011

industry feared that export controls could be more severe in the future (22%).[5]

Relying on sources outside Europe creates complications for European industry in terms of communication and quality control, and also introduces more risks with respect to security and foreign threats. Reliance on components developed outside Europe results in longer lead times, programme cost overruns and schedule delays, and the potential non-availability of high end technologies for some European satellite manufacturers. The lack of access to leading edge technologies inevitably produces limitations on the performance and scientific output of European payloads, which directly affects the competitiveness of European industry.

Dependence is often associated with insufficient technology maturity and lack of incremental development of critical technologies.[6] By critical technologies is meant those technologies that *enable Europe to be capable of responding to its own needs in space and to remain competitive in the global market.*[7]Amongst these technologies, those on which Europe is most dependent for foreign supply are the EEE-components. The impact of this dependence can be easily understood when considering that 40% to 70% of the hardware procurement costs of space equipment are for EEE-components.[8] On a standard ESA satellite programme, more than half of the EEE-component procurement costs go to procurement of components from the U.S. that are under export control restrictions.[9] In 2006 the average European spacecraft mission employed just 25% of European-sourced parts (the remaining parts were bought mostly from the U.S. but also from Asia). In 2010 this has risen to 35%, and today to 50%.[10] Even if some technology gaps between Europe/USA/Asia have been reduced - and are supposed to be reduced more – the gap is much too big for Europe to take an independent approach and rely only on European-sourced technologies. Although the European supply chain is still increasing, the gap is too big to achieve technological self-reliance and independence.

For these reasons, Europe must establish a Non-Dependence strategy. While *independence* would imply that all needed space technologies are developed in Europe, *Non-Dependence* refers to the possibility of Europe having free and unrestricted access to any required space technology.[11]

As a general topic in terms of technology Non-Dependence is increasingly a priority for European policies and institutions. Even though some European nations have achieved independence in some very specific sectors and technologies, an overall strategy at European level is still lacking.

Currently in Europe, national space agencies are organised with very different specific goals and regulatory frameworks and are committed to fulfilling mainly strategic, economic and industrial objectives in the interest of their own country rather than of Europe as a whole. Within each of the ESA Member States, the approach to space activities covers a broad spectrum that finds a first point of convergence only in the activities of ESA itself.[12] The coordination of efforts across all European actors in space-related activities, first and foremost at intergovernmental and institutional level, is therefore a prerequisite for maintaining and improving the position of Europe and reducing its foreign dependence**.** In a broader context, European fragmentation at the political, market and regulatory level is a further obstacle to the implementation of a coherent, overall and effective strategy.

The mechanisms driving technology in the direction of short-term competitiveness and commercial programmes are profoundly different from those driving technology for strategic reasons, such as Non-Dependence. In fact, the latter require comprehensive and long-term preparation and planning.[13] Naturally, strategic actions will also have an impact, directly or indirectly, on the space economy and commercial activities, yet the return on investment will materialise only late in the process. Funding should be allocated taking into account the long term, considering that the revenue from some of the activities will come only a decade later, and might not be directly traceable to the original investment.

This is why institutional support for R&D in space technologies and for long-term space

[5] ESTMP 2011
[6] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[7] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[8] *EEE Non-Dependence,* presentation by Mikko Nikkulainen, ESCCON Conference, March 2013, ESA-ESTEC
[9] ESTMP 2011
[10] *ECI Update,* Mikko Nikkulainen, 2015
[11] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[12] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[13] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

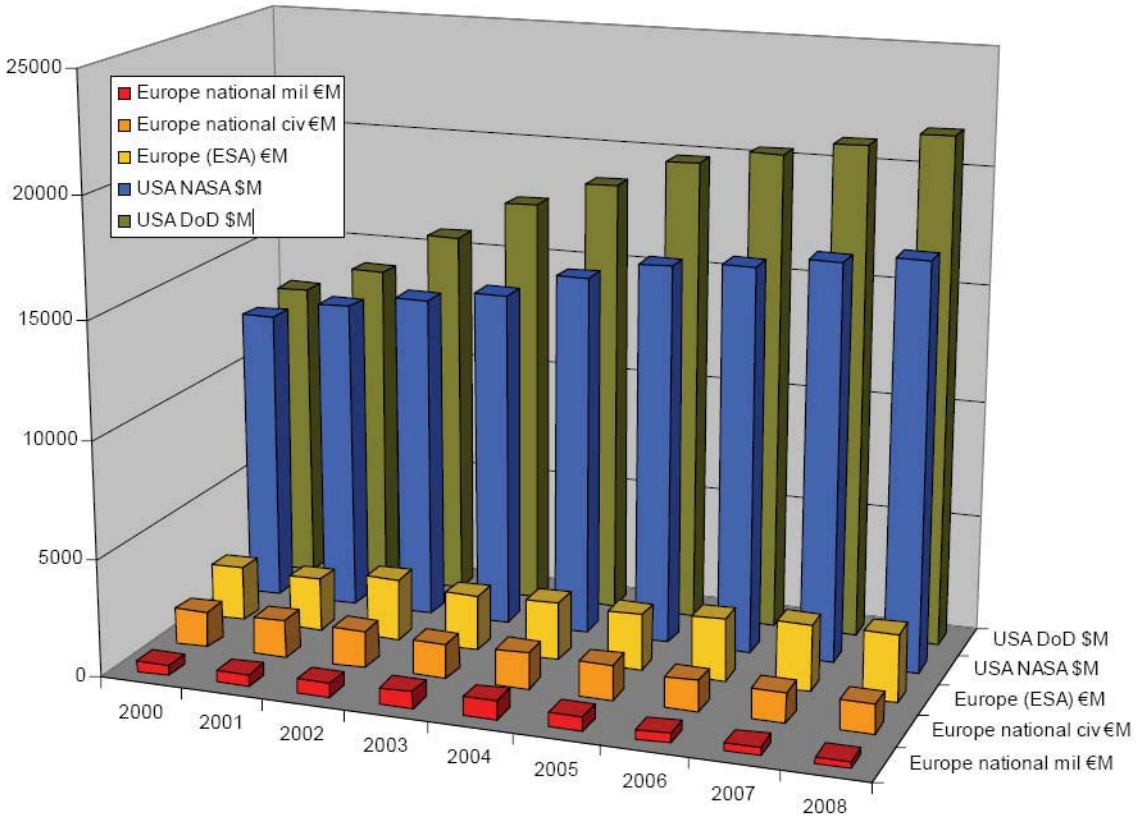


Fig. 1: USA and Europe funding for space activities. Figure from: *FP7-Space: R&D activities in support of European microelectronics enabling technologies,* presentation by Richard Gilmore, ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011

programmes is imperative. The high costs and high level of risk inherent in innovation and technological development mean that only public funding will do the job.

Funding should be devoted to enhancing a broad technological base and promoting innovation from the product concept to the later stages of the product life cycle, directly leading to the commercialization phase. End-to-end funding is fundamental to guaranteeing the implementation of any strategic plan.[14]

Most of the gaps in the development of cutting edge technologies are due to insufficient funding. U.S. funding for space programmes is about ten times that of Europe (see Fig. 1).[15] This enormous disparity is the basis of the leadership role of the U.S. in technological innovation and the global space market.

Any Non-Dependence strategy must be centred on critical and enabling technologies. Their availability and maturity level and impact on European commercial activities and security must be analysed and considered.

The general guidelines of innovation policy in this field must aim at the reduction of the dependence level of European space technologies through double action. On one side, ensuring unrestricted access from suppliers abroad and, on the other hand, encouraging and enhancing the development of building blocks and components from European suppliers, in order to strengthen the European supply chain. Such a parallel approach is inherent in the definition of Non-Dependence, and must be pursued to guarantee security of supply and maintain a solid position of Europe in the global scenario.

Alongside long term prospects for breakthrough concepts and avoidance of technology gaps, special attention should be given to the later stages of the technological process. It is necessary to improve European design tools and invest in the maturity level of technology, in order to promote the development of reliable standard equipment, especially for commercial use[16] (see paragraph 1.5). Dependence is often associated with insufficient

[14] ESTMP 2011

[15] FP7-Space: R&D activities in support of European microelectronics enabling technologies, presentation by Richard Gilmore ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011

[16] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

maturity of technologies and lack of incremental technology development. The later stages of development are often not covered by funding, and this produces a negative impact on European competitiveness and commercial activities.

Technology readiness is fundamental. It is the guarantee that the European space industrial base is able to supply the technologies required by European space programmes. Improving readiness and efficiency will lead to Non-Dependence and competitiveness of technologies and products.[17] The creation of harmonised and reliable standards, in the very fragmented European industrial and technological context, is crucial for the achievement of Non-Dependence goals. For space, this is even more complicated considering that building blocks for different space systems and applications are characterized by a lack of standardization: telecommunications, Earth observation, exploration etc… need different technical solutions and different satellites to be developed. Moreover, the technologies need long qualification processes and high investments, compared to the European space market size.[18]

The European space market is limited, not well integrated and currently, because of the economic crisis, additional industrial investment cannot be expected. Furthermore, technology cycles are becoming shorter, while time to market still remains much too long. Institutional actions jointly with stakeholders and customers should be undertaken to provide sufficient anticipation and aggregation of demand, optimizing every effort. Access to the most advanced technologies, components and materials is a condition for cost effective institutional missions and for the commercial competitiveness of the European space industry. In this respect, it is important to take into account the new and emerging spacefaring nations with major technological investments in the global context.[19] For Europe, this could represent a threat to its competitiveness on the worldwide market, especially for low cost technologies. Nevertheless, it could become a notable opportunity, given that the size of the European market is so small that Europe needs export of its products and could benefit strongly from new markets as emerging countries enter the space scene. This requires the support of a well-defined regulatory framework for import-export of sensitive technologies, as well as the definition of guidelines of foreign policy and international relations strategy.

An overall strategic approach should take into account the possibility of exploiting synergies with research and development in non-space sectors, in order to develop spin-in strategies and co-funding activities. First among these is the defence sector, which is contiguous to and shares many characteristics with the space sector, namely very demanding requirements, limited volumes, and cutting edge technologies. Europe's limited commitment to security-related space activities has led to technological deficiencies because of insufficient investments in some areas.[20]Co-ordination and an increase of efforts in space-related activities is a prerequisite for improving the strategic place of Europe in the world and for reducing dependency on the leader in the sector, the U.S. Increasing efficiency requires reinforcing complementarities and collaborations amongst stakeholders (both space and non-space sectors) to avoid duplication in each element of the value chain. To achieve a Non-Dependence strategy a multilateral and multi-stakeholder approach is necessary, at both the cross-sectoral and trans-national levels, while also strong coordination between the European Commission and ESA is crucial in this respect.

From this first general overview it emerges that security of supply is not just a technological issue - it requires strong and tailored political action to drive European technology policy. Europe must enhance technological investments, foster harmonisation between European actors (and, in this respect, also benefit better from relevant EU sector-based policies), and foster international relations for mutual exchange, commercial agreements and other forms of collaboration. To manage all these different aspects, strong, shared and effective policy action is needed.

1.2 Dependency Situations

Dependency issues can be very different in risk and criticality, in relation to the programmatic importance of the technologies considered. They can arise in many different situations, due to technological, economic or regulatory barriers and obstacles.

The following is a list and brief description of the most recurrent ones:

[17] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[18] ESTMP 2011
[19] ESTMP 2011

[20] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

European Technologies Are not Available at All

This is the most critical case of dependence, with a higher level of risk when items and technologies are associated to single supplier sources. This kind of dependence is mostly associated with building blocks and components, relatively low value standard items that are not available because of high costs for the producer, starting with the investments necessary to develop relevant capacities and then with the volume of activities necessary to sustain these capacities on a competitive basis.[21] In particular, Europe lacks some *Electrical, Electronic and Electromechanical* (EEE) components, some raw materials such as rare earths, and some advanced materials (as described in Chapter 3 and below).

European Technologies May Be Available But not With the Appropriate Level of Maturity, Reliability or Cost.

This is a critical issue principally for operational and commercial systems. Items affected by this kind of dependence are produced in Europe, but at a low level of development with respect to global market trends and needs. High performance complex components, needed to give a competitive advantage to space products, are often not sufficiently upgraded. In many cases, an advanced and competitive level of maturity would be achievable by European stakeholders, but it would require such high costs and lead times that it becomes more convenient for industry to buy the improved technology from suppliers abroad.

Most funds currently allocated to technology development programmes in ESA, national agencies and more recently the European Commission and EDA are devoted to the early stages of the technological process (from early concept investigation to breadboard models). Later stages of development, including all validation and demonstration phases, where less innovation and higher costs occur, are still inadequately covered.[22] For commercial applications in particular, *In-Orbit-Validation* (IOV) is a very fundamental step (as will be described in paragraph 1.5.1) and Europe would be well-advised to strongly support its capabilities in this respect.

European Technologies Need to Be Replaced because They Are Absolete, no Longer Available or Do not Meet New Legal Standards

Obsolescence is an extremely significant problem to deal with in the innovation process because of very rapid technological developments and the high competition in the space market. To manage medium/long term obsolescence, the right balance between innovation and product maintenance has to be found obsolescence affects technology and product availability and may occur as a consequence of many different conditions:[23]

- A technology is no longer produced by the supplier
- The process of deciding, funding and finally commercialising new developments is too slow with respect to global trends and market (see also the point *time to market*)
- Some components and materials need to be replaced because new European legislation prohibits their use.

  In particular, two new EC environmental regulations have an impact on space technologies and systems. These are:

  REACH: *Registration, Evaluation, Authorisation and Restriction of Chemical Substances*. EC Regulation No 1997/2006 which entered into force in June 2007, with a phased implementation over the following decade. REACH addresses the production and use of chemical substances and their potential impacts on both human health and the environment and applies to all chemicals imported or produced in the EU, including some substances that are contained in objects.[24]

  RoHS: *Restriction of Hazardous Substances,* Directive 2002/95/EC was adopted in February 2003 by the European Union and took effect on 1 July 2006 to be enforced and become law in each member state. This directive restricts the use of six hazardous materials in the manufacture of various types of electronic and electrical equipment.[25]

[21] ESTMP 2011
[22] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[23] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[24] http://en.wikipedia.org/wiki/Registration,_Evaluation,_Authorisation_and_Restriction_of_Chemicals
[25] http://en.wikipedia.org/wiki/ROHS

Strong Limitations on European Industry in Accessing Needed Technologies from Foreign Suppliers (Strategic Export Regulations)

The large majority of space items is dual-use. As with all sensitive technologies, this puts them under special attention and restrictive rules, in particular referring to export regulations. Thus, export restrictions may apply to basic technologies, products, materials and processes.[26] The prime exporter of space components and systems to Europe is the U.S., which applies extensive and strict rules through many regulations on the exportability of sensitive technologies. Exports from the U.S. are governed primarily by the *International Traffic Regulation of Arms* (ITAR). ITAR export regulations may prohibit the availability of a technology or, in any event, create procurement delays and put European industry in a further situation of dependence towards US policy instability.

Time to Market

Time to market is one of the main criticalities for European industry in every sector and has a huge impact on space technology development and Non-Dependence. It is a common situation that the production and/or upgrading and commercialization of some European-made technologies requires so long a time that the item becomes obsolete soon after marker introduction. This encourages European stakeholders to get the necessary technologies and components from suppliers abroad, where the whole process is faster and the expense is lower, in order to avoid obsolescence and staying competitive in the global market. As a consequence, this puts them in a condition of dependency on supplier countries. The innovation timeframe of the commercialization phase of technology development has to become more compatible with market dynamics.

Excess of Fragmentation in Europe /Un-Harmonised Regulations

The very long timeframes needed for the process of technological commercialisation can be due to different factors. One of the more relevant is excessive fragmentation in Europe. The political fragmentation of the EU has direct implications for the achievement of an innovation policy for Europe as a whole, in particular with reference to regulatory, industrial and market aspects.

By the very nature of the EU, the many different regulatory systems at national level are still predominant with respect to European directives and regulations, and are often not harmonised. Moreover, despite the many actions toward the establishment of a single market in Europe, European markets are still strongly fragmented, and the situation is even more complex in the high-tech field because of security concerns.

Both regulatory and market fragmentation in the EU create obstacles for transnational collaboration and crucial procurement delays that make the whole European system move like a slow machine with respect to its competitors on the global market. In order to promote and improve technological advancement and to guarantee the achievement of Non-Dependence for Europe, this is one of the main issues to be dealt with. In particular, the establishment of well balanced and harmonised regulations on export controls, both with countries outside the Union and inside European borders, is highly relevant for the more general purposes of strategic opportunities, competitiveness and Non-Dependence. In order to boost the position of Europe globally and allow single nations to collaborate in in a very effective way, all this has to be taken into consideration.

It appears evident that Non-Dependence is a very complex and overarching topic and, as such, it must be analysed from many different perspectives.

1.3 Levels of Dependence Criticality

Dependence situations can be very different in risk and criticality in relation to the programmatic importance of the technologies considered. In order to have a universal scale to refer to, industry has defined three levels of dependence criticality:[27]-[28]

- *Level 1:* No risk of mid-term unavailability. The technology is available at reasonable conditions in the short/medium term for European programmes. A potential risk could be perceived in the longer term. Monitoring the situation is required to anticipate potential issues.
- *Level 2*: Long-term availability is not guaranteed. Access to the technology is

[26] ESTMP 2011

[27] *FP7-Space: R&D activities in support of European microelectronics enabling technologies,* presentation by Richard Gilmore ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011

[28] *Industrial Reality of Technological Dependence,* Evert Dudok, presentation from the conference *Critical Space Technologies for European technological Non-Dependence,* Brussels September 9th, 2008

guaranteed in the short to mid-term but access in the longer term is at risk.

- *Level 3:* Access to needed technology is seriously restricted. Level 3 is further split into three levels of seriousness:

  3.1: A technology is available in Europe but it is not competitive for performance, maturity or cost reasons.

  3.2: A technology is not available in Europe and its access is restricted. This is the general case of export control rules.

  3.3: A technology is available but the offer is limited. This is critical if space competes with bigger customers for access to resources.

The following figure describes, in the form of a criticality tree, the procedure used in ESA to assign critical levels to technologies.

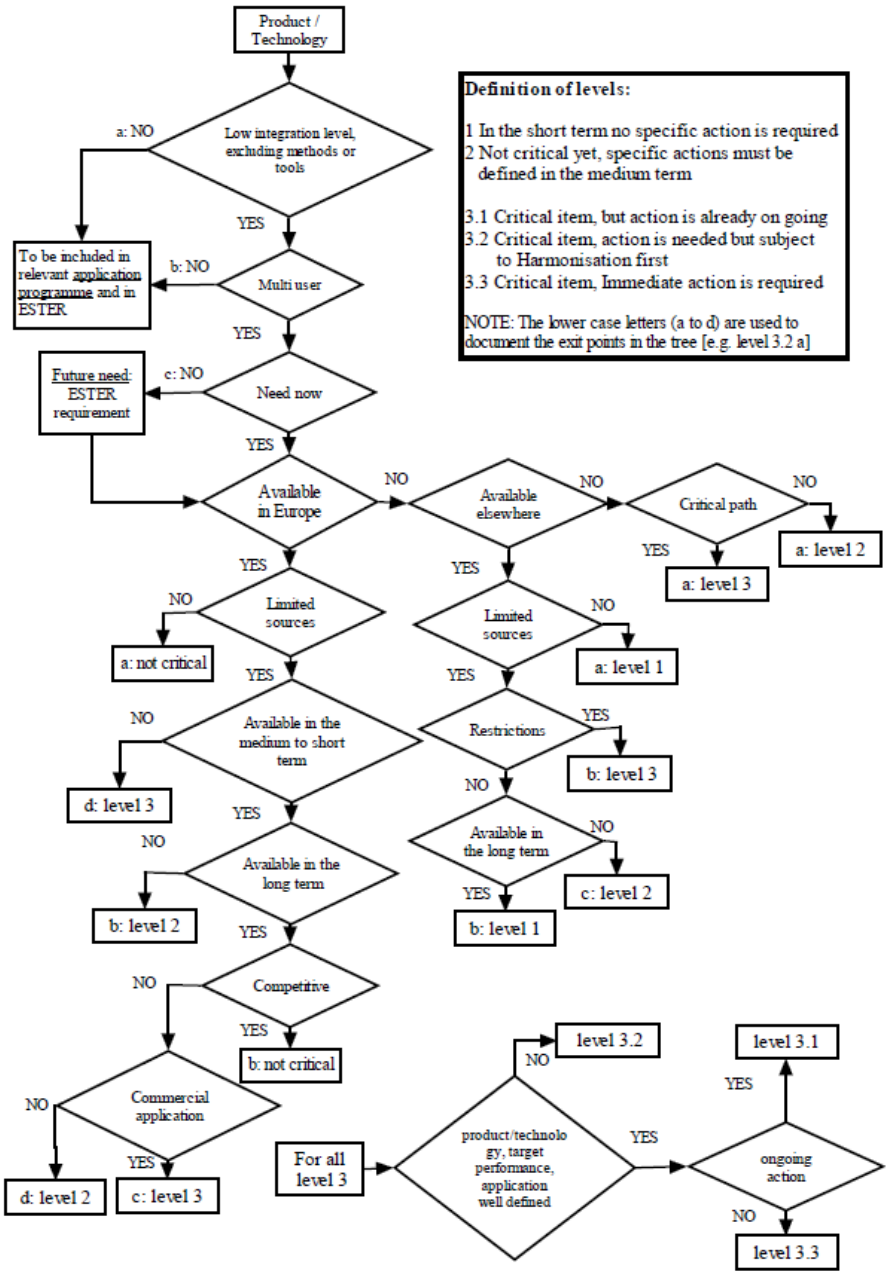


Fig. 2: ESA criticality tree. This scheme shows the method used in ESA to establish critical levels for the considered items. Figure from: *2012 analysis of the Non-Dependence situation for critical space technologies in Europe, ESA industrial Policy Committee, Information Note, Appendix A.*

1.4 General Context

1.4.1 Global Supply Chain

The supply chain encompasses the design, development, testing and manufacturing of components, systems and facilities for access to and operation in space. All these different phases in the creation of an integrated technological system are strongly connected. This is why even a small disruption or damage in the provision of a single component affects the entire production line. Dependence situations that may arise not only from technical problems but also from industrial and commercial conditions or export restrictions have an impact far beyond the value of the single items affected on the overall supply chain.

Today, for any complex technological system, and of course for space, the supply chain has a global dimension. This increases potential risks and uncertainties due to political instability in countries involved. In the same way, any regional disaster affecting a single nation has global repercussions (as for example the Japan crisis of 2011 after the earthquake and tsunami). The globalized nature of the supply chain leads to the establishment of Non-Dependence strategies, as previously defined. Security of supply is a driver for any technology strategy, and must be implemented during the whole process of innovation. For this purpose, the full traceability and transparency of the product life cycle and supply chain levels are fundamental.[29]

The European space industry is proficient along most of the supply chain, *except at the lowest levels* with basic supplies, such as EEE components and advanced materials. This is, however, a very critical condition, since technological dependency on components or raw materials makes Europe more vulnerable to political changes or natural disasters in other regions. In fact, it is clear that the lower the level of the supply chain is on which a country is dependent, the higher is its vulnerability.

As mentioned above, at present only 50% of EEE-components on an average European space mission are produced in Europe. At the basis of modern electronics – and of the supply chain - there are the semiconductors, which deserve special attention when talking about EEE-components, building blocks and advanced materials. Semiconductor-based electronic components include transistors, solar cells, many kinds of diodes, light-emitting diodes (LED), photo-diodes, etc. Semiconductors are used to create integrated circuits, which are at the foundation of any complex electrical and electronic device. The process of fabrication of integrated circuits is a multiple sequence of photolithographic and chemical processing steps, through which electronic circuits are gradually created on a wafer of pure semiconducting material. A large number of elements and compounds have semiconducting properties, including pure elements. Silicon is almost always used, followed by germanium, but many compounds are produced for specialized applications (such as gallium arsenide, silicon carbide and a number of organic compounds). The enhancement of understanding of semiconductor materials and fabrication processes is key for the progressive increase in the complexity and speed of electronic devices and, with them, of increasingly sophisticated integrated systems. Looking at the space sector, the high level of complexity and specification of technological devices and the great number and differentiation of applications necessitate very close cooperation with the semiconductor industry from the early stages, to achieve effectiveness and higher performance.[30] Non-dependence strategies aim at making such cooperation possible and effective.

High-Tech Components are Imported from U.S.

The U.S. is the first and principal exporter of hi-tech devices for space systems to Europe. With respect to electronic components in particular, Europe currently imports from the U.S. about 60% of the total number of EEE parts present on an average European mission.

As already stated, space-related U.S. technologies fall under the *International Traffic in Arms Regulations (ITAR).* This is a set of government regulations that controls the export and import of sensitive defence-related items listed on the *United States Munition List (USML).* The Department of State Directorate of Defence Trade Controls (DDTC) interprets and enforces ITAR. ITAR regulations prescribe that information and materials connected with defence and military technologies (and so, on the USML) may only be shared with U.S. persons unless authorization from the Department of State is received or a special exemption is given. This is done to safeguard U.S. national security and foreign policy objectives.

[29] *EEE Non-Dependence,* presentation by Mikko Nikkulainen, ESCCON Conference, March 2013, ESA-ESTEC

[30] http://en.wikipedia.org/wiki/Semiconductor_material, http://en.wikipedia.org/wiki/Semiconductor_device_fabrication

The U.S. Munitions List changes over time. The redefinition most relevant for space occurred in 1996, when technology pertaining to commercial satellites, until then regulated by the Department of Commerce, was shifted from the Commerce list of dual-use items to be put again under the USML. In this way, most space technologies became subject to higher regulatory restrictions and protective measures. This happened as a result of the launch failure of China's Long March Rocket in February 1996 with the Space Systems/Loral Intelsat 708 .The Department of State charged Space Systems/Loral with violating the *Arms Export Control Act* and the ITAR, and opted for a renewed hardening of controls.[31] There is an open debate between the Department of State and industries and academia regulated by ITAR on how such regulatory restrictions can damage U.S. businesses and higher education. The security benefit to the nation should be weighed against the economic impact.

For Europe, ITAR export regulations often create procurement delays and exposes European industry to U.S policy changes.[32]

Microelectronics and Semiconductors are Mostly Imported from Asia, Mainly from Japan.

Japan plays a major role in the global supply chain both as a supplier of parts and as a producer of final products. Japan's produces more than 60% of the global supply of silicon wafers.[33] Japanese equipment and material suppliers are leading players in the global semiconductor industry. It is estimated that Japan-headquartered equipment companies collectively capture about 35% of global semiconductor industry spending per annum. Beside wafers productions, chemical and other material suppliers in Japan are market leaders in the manufacture of advanced chemicals, packaging resins and packaging substrates. However, the importance in Japan of the semiconductor industry has receded in recent years (less than 21% of total production, down from 28% in 2001) as more production has shifted to South Korea, Taiwan and China.[34]

Rare Earth and Material Dependencies on China

In early 2010, Chinese firms accounted for 97 percent of rare earth oxides production and a large fraction of the processing business that turns these into rare earth metals and products. In September 2010, during a maritime border dispute with Japan, China apparently embargoed exports of rare earth oxides and metals to Japan (Japanese companies insisted the embargo was real, even as the Chinese government officially denied it).[35]

Japan, the United States and several European states, concerned about Chinese embargo and also motivated by expected increases in demand, started to invest in the opening of rare earth mines and building new focused technologies and processing capabilities. Other investors were moving ahead on mines in Canada, South Africa, and Kazakhstan. The major investments, made by Molycorp in the United States and Lynas in Australia and Malaysia, started delivering non-Chinese rare earths to global markets in 2013. This increase in non-Chinese production and processing capacity has led to reduced demand for some of the most crucial rare earths elements from China.

Although Chinese producers still contribute the great majority of supply, surveying trends in rare earth supply, demand, and processing, leads to the expectation that China's supremacy in the rare earth market will diminish. Moreover, in the current market, China appears technically slower, using, for example, old, environmentally destructive extraction technologies, while Japan and Europe are investing in more innovative and sustainable technologies. Although this expanded supply has alleviated China's dominance, the problem on rare earth material still remains.[36]

To guarantee its security and to be more competitive, Europe needs to develop a stronger and enhanced European supply chain. Nevertheless, considering the technological gaps of the EU, this cannot be enough, and the global supply chain will still be necessary. So, Europe needs to balance its efforts and investments in technological advancement with its ability to engage in partnerships with other countries. This implies overall strategic action.

Considering the important role of Asia as a potential partner that shares the need to emancipate itself from ITAR restrictions and the American monopoly, a brief description of the latest space-related technological achievements of some emerging and mature Asian countries is set out below.

[31] *A short history of export control policy,* Ryan Zelnio, The Space Review , January 9, 2006
[32] *A short history of export control policy,* Ryan Zelnio, The Space Review , January 9, 2006
[33] fpc.state.gov/documents/organization/159785.pdf
[34] http://theautomaticearth.blogspot.it/2011/03/march-21-2011-stable-in-sense-of.html
[35] Rare Earth elements and National Security, Eugene Gholz, Council on Foreign Relations, October 2014
[36] Rare Earth elements and National Security, Eugene Gholz, Council on Foreign Relations, October 2014

1.4.2 Space Programmes in Asia

In recent decades several Asian nations have emerged as space actors. As Bertrand de Montluc states the development of space activities in Asia is motivated by three aspects: the pragmatic, symbolic and diplomatic[37] With regard to the pragmatic aspect, China, Japan and India are giving priority to the development of infrastructure in telecommunications and Earth observation to meet socio-economic needs. Regarding the symbolic level, the nation which cares most is China, which, for instance, has achieved to put humans in orbit. In this way, China aims to create confidence in the quality of its home-made high-technology and to ensure regional and international influence at higher levels with respect to its competitors Japan and India. On a diplomatic level, the main goal is to make space services available to third states that have smaller financial resources for space and suffer from ITAR restrictions (e.g. China with Brazil, Venezuela, Bolivia and Pakistan).

Another characteristic of space policy in Asia is that objectives are not integrated, and individual nations carry out their own plans without generating programmatic joint actions. In fact, although some regional forums on the use of space application have been established, competition is still vastly predominant, and Asian individual states seek bilateral cooperation with European and Western countries rather than collaboration between themselves for the creation of an integrated, regional power. Emerging and new space countries are obtaining greater regional and international weight and reliability. They represent new opportunities for export sales and scientific cooperation, as well as becoming new players for the dissemination of satellite applications. For this reason, the space community should welcome these developments, especially Europe, with the small size of its space market and the importance of exports. On the other hand, their emergence and increasing degree of maturity will also produce commercial competition and a change in the regional and international strategic equilibria. Moreover, the growth of Asian countries in space includes military power, with many implications for security in the global context (China and India, for example, are heavily investing in the utilization of space systems for defence and security. Japan is a special case, being closely associated with U.S. but suffering, as other countries, from ITAR export controls).

[37] *Asia and Europe facing the technological revolution*, Bertrand de Montluc, nova Science Publisher, 2013

1.5 Technology and Innovation Policy

1.5.1 Overall View on Space Technology, the Innovation Process and the European Actors

By its very nature, space technology is constantly evolving, showing increasing complexity and many peculiarities.

Any spacecraft is the result of components, building blocks, equipment, sub-system and system integration. At system level, the main technology drivers are, on the one hand, the increase of functions, power, size and capabilities (e.g. large Geo satellites) and, on the other hand, the reduction of mass, size and power consumption (e.g. formation flying/constellation concepts, nano-satellites). At equipment level, technology drivers are the miniaturisation of mass/size, functional hybridisation and power and thermal aspects.[38]

Since it integrates a broad range of technology domains (propulsion, mechanisms, electronics, software, materials, ...) with a common set of requirements but very different specifications and purposes, space technology is highly diversified.[39] The lack of standardization is a typical feature of space applications. Telecommunications, Earth Observation, exploration and other space specialisations need different technical solutions, and so different satellite types must be developed.[40]

Another relevant issue is the dual-use (military and civil) and thus sensitive character of space technology. This has made innovation and commercialization of the elements of most space systems subject to very special attention and also resulted in many obstacles to be overcome, for example at regulatory and security level. This has a notable impact on technology policy actions (see paragraph 1.8)

Furthermore, space technology is associated with very high risk levels, for the harsh space environment and so demanding performances and cutting-edge technical solutions are required.

For all these reasons, the space sector needs innovation and constant control of technology

[38] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace, Annex A, Space technology characteristics.
[39] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace, Annex A, Technology diversity
[40] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

evolution throughout the whole product life cycle. Innovation, heritage and obsolescence are key issues for space systems and technology that need specific attention. Heritage is an essential prerequisite, but should not hinder necessary innovation. Obsolescence must be constantly controlled and can arise for many reasons, as explained above.[41]

A technology policy for space should optimise technological choices balancing maturity, reliability and predictability with highest performance and innovation. This is the prerequisite for the achievement of any strategic goal in space and competitiveness in the global market.

Finally, as a consequence of all these features, space technology is capital intensive. Development of space programmes requires huge investments in industrial equipment, including test and integration machinery, software, design and modelling tools, protocols development and maintenance, while also considering the high level of risk and the performance requirements.[42]

Currently, the main actors for European space technology are:[43]

- ESA (200-250 M€/year available for technology development)
- Industry (150 M€/year investment to support technology and competitiveness)
- National space agencies (100 to 140 M€/year)
- The European Commission (FP7/Space/Space Foundations, 10 M€/year)

Public support has been absolutely necessary because of the high costs and risks involved in technology development, and the comparatively low returns from domestic commercial and institutional markets when compared with the U.S., Europe's main competitor, and considering the increasing funding of the space sector by governments of emerging space players such as China, India and Brazil.[44] European national governments have been responsible for the definition of a Space Policy for Europe, mostly through their role and participation in ESA programmes, but also through activities conducted at national level.

Comparing the needs and costs of production, availability assurance and maintenance with the amount of public funding specifically devoted to space in Europe, the necessity for enhancement, also with respect to other space powers, first of all the U.S., is evident. However, the financial crisis affecting Europe makes any budget augmentation particularly hard to obtain. This means that to boost the development of the space sector, stakeholders are forced to make use of any EU, national and regional initiatives that could enhance space activities in a synergistic way.

On the institutional side, funding should be harmonised with the technology plan of each national space agency and be tailored for the longer timeframe. Funding timeframes tend to be short and thus inadequate to cover strategic actions. To give a general example, the Multiannual Financial Framework (MFF) of the EU covers periods of seven years. Although the timeframe of the Framework Programme for Research and Innovation has been raised from five to seven years, in order to harmonise the two programmes and improve effectiveness (with the new *Horizon2020* programme) for space activities, this is still inadequate for true strategic planning in the technology field.

The innovation process for the creation - or upgrading - of a new technology is a long process characterized by many and very different stages. For each stage, an adequate level of maturity and readiness should be achieved before passing to the next one. Technology readiness is the guarantee that the European space industrial base is able to supply the technologies required by European space programmes.[45]

A *Technology Readiness Level (TRL)* scale is used to assess the maturity of a technology during its development phases in a clear and universal way. Levels range from 1 to 9, referring to the different phases of the development of items in innovation programmes, from proof of concept to commercialization. TRL 1-3 covers fundamental innovation, basic technology research and feasibility analysis; TRL 4-6 concerns the development, qualification and demonstration phases; TRL 7-9 relates to the system/subsystem development, in orbit validation (IOV) and all phases oriented to end-users.

The TRL levels for space technologies were initially defined by NASA and have now been officially adopted by European space stakeholders.

[41] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[42] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace, Annex A, Space technology characteristics.
[43] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[44] ESTMP 2011
[45] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

In the table below, technology readiness levels with their specific definition are described in detail.

Level	Definition	Explanation
TRL 1	Basic principles observed and reported	Lowest level of technology readiness. Scientific research begins to be translated into applied research and development.
TRL 2	Technology concept and/or application formulated	Once basic principles are observed, practical applications can be invented and R&D started. Applications are speculative and may be unproven.
TRL 3	Analytical and experimental critical function and/or characteristic proof- of-concept	Active research and development is initiated, including analytical / laboratory studies to validate predictions regarding the technology.
TRL 4	Component and/or breadboard validation in laboratory environment	Basic technological components are integrated to establish that they will work together.
TRL 5	Component and/or breadboard validation in relevant environment	The basic technological components are integrated with reasonably realistic supporting elements so they can be tested in a simulated environment.
TRL 6	System/subsystem model or prototype demonstration in a relevant environment (ground or space)	A representative model or prototype system is tested in a relevant environment.
TRL 7	System prototype demonstration in a space environment	A prototype system that is near, or at, the planned operational system.
TRL 8	Actual system completed and "flight qualified" through test and demonstration (ground or space)	In an actual system, the technology has been proven to work in its final form and under expected conditions.
TRL 9	Actual system "flight proven" through successful mission operations	The system incorporating the new technology in its final form has been used under actual mission conditions

Fig. 3: Technology Readiness Levels. From: *Space technology in FP8/CSF*, Technology Policy Working Group Position Paper, Eurospace. Source: The TRL Handbook, issue 1 rev. 6 – in preparation as an input for ISO WG

As mentioned earlier, most funds associated with technology development programmes in national agencies, ESA, EC and EDA, target the initial stages of technology development, namely from early concept investigation to breadboard models. Later stages of development, including all validation and demonstration phases associated with less innovation and higher costs, are still insufficiently covered.[46] Current and future policies should enhance the commitment toward technology maturity aspects, a fundamental issue for reliability and competitiveness in the global market. As already mentioned, the lack of maturity of accessible components is one of the main causes that give rise to dependence situations. To translate advanced technologies into products and services it is necessary to increase support for development especially in what is generally called the "*valley of death*", that is the part of the product life cycle between the end of the investigative phase and commercialisation (broadly, TRL 4-6). For this purpose, it is imperative to include funding for qualification up to at least TRL 6-7.[47] This is necessary and should be pursued at all levels of complexity, both for equipment/integrated systems and for components/building blocks.

Building blocks are the first level of integration and must be isolable, have a clear function and be complete with respect to this. They have precise interfaces for integration at higher levels. Building blocks are developed to TRL 6 or beyond. Being at the lower level of integration and of the supply chain, components and building blocks are crucially important and deserve special attention. The Joint Task Force established by EC, ESA and EDA for the implementation of a European Non-Dependence strategy, recognising the decisive importance and European weakness in this specific field, chose to concentrate on

[46] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[47] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

components and building blocks access and upgrade in its first urgent actions (see Chapter 2)

The space sector benefits from technologies with proven or even promising concepts (TRL 2-4) coming from other market areas and other fields of applications (military and defence, for example, but also less close domains). Non-space technologies are upgraded by incremental developments in order to bring them to TRL 6-9 for use in space programmes (some examples are CCDs[48], memory, FPGAs[49], solar cells, batteries, MEMS/MOEMS[50], materials composites, etc…). In this sense, a true spin-in strategy is necessary. This is mostly relevant for components and building blocks.

An adequate technology strategy must be able to bring the most promising and affordable technologies (at component, building block, equipment and system level) to the appropriate maturity level for risk-free implementation in European programmes. This requires that technologies have passed a long qualification and evaluation path, ending with in orbit validation.[51]

Just as pursuing maturity - especially at the later stages of process development - is mandatory to achieving Non-Dependence, failing to pursue technology innovation definitely ends in dependence situations.[52] Technology strategies for innovation, and subsequently for Non-Dependence, must take into account this fine and fragile balance.

Alongside these general guidelines, a key issue for effective technology strategies is the definition of final specifications and their optimization. This is crucial to enable a unified perspective at European level and a cooperative effort and joint activity at research and industrial level, both between European nations and with non-European countries. For this purpose, the definition of very precise specifications needs to be accompanied by an overall standardisation process of the different items and procedures. This must involve all European actors to be able to establish the basis of a unified strategy in the fragmented European landscape. In the present scenario of open innovation and the knowledge society, standardisation is needed to allow true, joint initiatives between European countries and to guarantee their effectiveness and, consequently, the reliability of European technologies. This is very significant also with respect to non-European partners or customers, in the light of current or future international collaborations and agreements.

As previously observed, space technology is segmented from a product point of view, with technical differences between families of products that are driven by different sets of requirements.[53] Europe needs to establish a coherent, shared product policy to link technology evolution to user requirements and to procurement programmes in an optimised fashion.

Currently, ESA is responsible for a range of space product policies. In particular, in the ambit of ESA technology management, user-driven and utilisation-oriented products are defined for European programmes and companies to be competitive in the world market; technology development programmes and processes are targeted to support the whole product development cycle, from concept to qualification, including in orbit validation, and processes and tools that allow releasing and maintaining a catalogue of such products are set up. ESA establishes measures to encourage investments by member states, and develops policies to promote the use and reuse of products in ESA and non-ESA space programmes. Moreover, equal opportunities are assured to all classes of firms, with particular regard to SMEs, in order to maintain a strong supplier base with repeatable products and processes.[54]

The need for a space product policy at European level is even more pressing considering that space technology programmes, with their demanding requirements and elevated costs, suffer from the small size of the Euro-

[48] *A charge-coupled device (CCD) is a device for the movement of electrical charge, usually from within the device to an area where the charge can be manipulated, for example conversion into a digital value*.
https://en.wikipedia.org/wiki/Charge-coupled_device

[49] *A field-programmable gate array (FPGA) is an integrated circuit designed to be configured by a customer or a designer after manufacturing – hence "field programmable".*
https://en.wikipedia.org/wiki/Field-programmemable_gate_array

[50] *Microelectromechanical systems (MEMS) is the technology of very small devices; it merges at the nano-scale into nanoelectromechanical systems (NEMS) and nanotechnology.*
https://en.wikipedia.org/wiki/Microelectromechanical_systems
Micro-Opto-Electro-Mechanical Systems (MOEMS) is MEMS merged with Micro-optics which involves sensing or manipulating optical signals on a very small size scale using integrated mechanical, optical, and electrical systems.
https://en.wikipedia.org/wiki/Micro-Opto-Electro-Mechanical_Systems

[51] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

[52] ESTMP 2011

[53] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace, Annex A, Space technology characteristics.

[54] *The ESA End-to-End technology Management Process*, presentation by Marco Guglielmi, Stephane Lascar, Edmund Williams, ESOC, May 2009

pean market, as mentioned. This is why space items considered for development are often those deriving from terrestrial non-space technologies, with bigger markets and more funding available than space.

This is the case, for example, for many microelectronic and nanotechnology items that have been developed and utilised in recent years. Space has taken advantage of the growing attention and fast technological and commercial evolution in these fields, fundamental for space because they are basic components of every integrated system. As observed above, working on space components is inherently a spin-in activity. The growth and widespread use of *Key Enabling Technologies* (KETs, see paragraph 1.7) and the attention that European institutions are devoting to them in terms of initiatives and funding because of their strategic, multifunctional and enabling value, could actually be a driver for space technology improvement, in particular for the necessary enhancement of the European supply chain for building blocks and components. For this reason, fostering the exploitation of synergies with research and development in other sectors through the accomplishment of a structured spin-in strategy will be crucial for space and innovation policy.

To take advantage of all these achievements more coordination is needed, with targeted actions, joint undertakings, strong synergistic efforts and technology transfer plans. Technology planning and coordination is mandatory, with the elaboration of long-term prospects for breakthrough concepts and associated technologies, as well as the involvement of all different actors in a structured multi-stakeholder approach.

Strategic actions such as Non-Dependence must be implemented over a long timeframe with the definition and monitoring of medium to long-term roadmaps. Coordinated multi-year work plans and yearly procurement plans must be harmonised with initiatives at national, regional and European level. The key competences already fully achieved by Europe should be preserved, and, to do this, capabilities should be encouraged, maintained and enhanced.

Technology is not just the sum of items and processes, but also the capability to manage them. Investing in human capital and maintaining competence must be an integral part of European innovation policy and any technology strategy. Through capability and competence, Europe can definitely achieve the Non-Dependence goals and remain a central actor in space.

1.5.2 ESA Technology Policy – Generic Programmes

As seen in the previous paragraph, technology and innovation need strong policy actions and, in particular in the fragmented European landscape, an effective coordination effort. The guidelines for space technologies and innovation policy are given by ESA, where the Member States, whilst also maintaining a degree of autonomy, participate in a joint strategy. For this reason in this paragraph ESA technology policy and management and general programmes are addressed.

R&D activities in ESA are managed in the Directorate of Technical and Quality Management (TEC). (Space technology R&D is defined as the part of R&D work between TRL levels 2-6).[55] ESA spends around 8% of its budget on direct technology R&D.

The planning and management of all ESA technology activities development is channelled through the *End-to-End (E2E) process*, while detailed programme formulation, funding and the interface with the respective Programme Boards are considered within each specific programme Directorate.[56] The main objective of the E2E technology process is to undertake well-planned technology development policy using long term plans and developments in a fully transparent, efficient and well-coordinated way. This is done according to user requirements - such as mission needs, industry competitiveness and Non Dependence goals – aiming to get technology ready at the maturity level required at each stage.[57]

In greater detail, the main technology policy goals pursued through the E2E process are:[58]

[55] ESTMP 2011
[56] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf
[57] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf
[58] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

- Integrating the different technology programmes within a common set of processes and goals
- Securing funding at the right time and for the right need or purpose
- Leveraging and reinforcing technology harmonisation and standardisation
- Setting a real product policy
- Developing a systematic monitoring and evaluation process of the technologies developed
- Exploiting synergies with non-space sectors, promoting spin-in strategies

The E2E process is structured in 9 specific Service Domains (SD), namely: Earth Observation, Science, Human Spaceflight and Exploration, Space Transportation, Telecommunications, Navigation, Generic Technologies and Techniques, Security, Robotic Exploration).[59]

Two key elements of the E2E process are: [60]

- *Technology Strategy and Long Term Plan (TSLTP).* It gives an overview of the Agency's strategic objectives, needs, priorities and future visions for the short, medium and long term. It covers a 10 year period, answering to ESA's overall strategy and Long Term Plan (LTP) and to the European Space Policy (ESP).
- *Technology Monitoring and Evaluation (M&E).* This measures and checks performance and results of each R&D activity with respect to the objectives. Moreover, it supports the definition of correction actions as necessary. Monitoring and evaluation have a complementary function: Technology Monitoring is the measure of the implementation of every technology plan with respect to the baseline plan, from the initial work-plan to the contract completion, while Technology Evaluation assess the achieved results and benefits of the completed technology R&D activities.

The management of the process is governed by two bodies: the *Directors´ Sub-Committee for Technology* (DSCT) and the *Technology Network* (TECNET). The DSCT is chaired by the Director of Technical and Quality Management and includes Programme Directors and the Director of Operations and Infrastructure. It is the inter-directorate body advising the ESA Director General on all subjects connected to technology. It is in charge of the overall supervision and coordination of the different technological activities and processes, and reviews and endorses the TSLTP and the proposals for Technology Plans at all levels.[61]

The TECNETs are nine working groups of senior experts from ESA technical directorates in charge of the service domains. TSLTP Activity Proposals are reviewed, selected and allocated to technology programmes by the respective TECNET Service Domains (SD) on the basis of prioritized technology requirements and roadmaps.[62]

The tasks of TECNET groups, each one in its specific ambit of interest, are:[63]

- To identify the user missions and associated technological requirements (through ESTER, see Chapter 2)
- To prioritize the technology requirements
- To promote, evaluate and select activity proposals for technology development according to technology needs
- To allocate activities to technology programmes
- To monitor and evaluate the implementation of the technology actions related to their domain

Technology is developed in ESA under general or several domain specific programmes with different temporal cycles of a 5-10 years

[59] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

[60] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

[61] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

[62] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

[63] *The ESA End-to-End Technology - Management Process*, Marco Guglielmi, Stephane Lascar, Edmund Williams, presentation at ESOC, May 2009, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_01_ESA_E2E_TM.pdf

time-span. The general programmes are the *Basic Technology Research Programme* and the *General Support Technology Programme.*

The ESA *Basic Technology Research Programme (TRP)* is the programme where new ideas are tried through proof-of-concept testing, covering TRL 1-3 of technological development.[64] Within the TRP, disruptive innovation is a special priority. The feasibility of a given technology is tested to factor in the availability of prospective and required innovations in the long-term planning of future missions. This provides Europe with solid technological capability and enables it to be at the forefront in the definition of new cutting edge space missions and applications with possible exploitation of products by industry on the world market.[65] This technological preparation reduces project risks in performance, time and costs. So, for ESA, TRP is the backbone of new ideas, contributes to scientific and engineering excellence and is a significant cost saver in the long term planning.[66]

Contributing to the TRP is mandatory for all ESA Member States. It is run on a three-year work plan and organised according to technology themes based on application areas. The TRP work plan is drafted by TECNETs with reference to the ESA Long Term Plan.[67] As the basis of most future development, the TRP is the only ESA technology programme that supports all Agency directorates across the whole spectrum of technical disciplines.[68] Its structure is divided into Service Domains, Technology Domains and Cross-Cutting Domains. In this last subject area, one of the main activities is the consideration of components, materials and micro-nano technologies.[69] In the Service Domains, the TRP allocates a third of its effort to *Generic Technologies*, either of use to multiple missions or else advanced basic technologies of common interest to all applications, such as component design, spacecraft propulsion or power generation. Relevant initiatives within the TRP include spin-in activities from non-space sectors through the *Star Tiger* scheme and *Innovation Triangle Initiative*.[70] The low TRL level of the technologies makes activities within the TRP very suitable for cross-cutting opportunities, synergies with different sectors and technology transfer. These are strongly supported in the ambit of the programme.

The *General Support Technology Programme (GSTP)* aims to convert promising engineering concepts into mature products, from individual components to subsystems and up to complete satellites. The GSTP bridges the gap between having a technology proven in its general features and making it ready for ESA and national space missions and the market, with a sufficiently high confidence level and a sufficiently low risk factor. Part of this programme takes as its subject technologies passed through the TRP process. It also responds to requests from industry for technical support and evaluation of products and it works on product and process improvements, for a targeted and flexible response to the needs of ESA programmes, Member States and European industry. Moreover, among its objectives is the enhancement of the number of commercially available European-made space-qualified parts and components.[71]

The GSTP is an optional ESA programme. GSTP activities cover all technical domains plus *Generic Technologies*, except Telecommunications (which has its own programme, ARTES, *Advanced Research in Telecommunications Systems*).[72] It has been in operation for two decades.

The current GSTP-5 operates on a five-year work plan, with a budget of ~ 350 million euros for the whole period.[73] It is based on four programme elements, organised on a voluntary basis:[74]

64

http://www.esa.int/Our_Activities/Technology/About_the_Basic_Technology_Research_Programmeme_TRP

65 *ESA TRP and GSTP*, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_03_ESA_Miro.pdf

66 *ESA TRP and GSTP*, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_03_ESA_Miro.pdf

67

http://www.esa.int/Our_Activities/Technology/About_the_Basic_Technology_Research_Programmeme_TRP

68

http://www.esa.int/Our_Activities/Technology/About_the_Basic_Technology_Research_Programmeme_TRP

69 *ESA TRP and GSTP*, http://www.egos.esa.int/export/egos-web/others/Events/Workshop/ESAW-workshop-2009/Day1-Session-4-1600-1745/S04_03_ESA_Miro.pdf

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http://www.esa.int/Our_Activities/Technology/About_the_Basic_Technology_Research_Programmeme_TRP

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http://www.esa.int/Our_Activities/Technology/About_the_General_Support_Technology_Programmeme_GSTP

72

http://www.esa.int/Our_Activities/Technology/About_the_General_Support_Technology_Programmeme_GSTP

73 *General Support Technology Programmeme*, presentation at ESA Thematic Information Day, Belspo, June 2012 http://www.belspo.be/belspo/space/doc/euPolicy/2012_06_25/GSTP-6.pdf

74

http://www.esa.int/Our_Activities/Technology/About_the_General_Support_Technology_Programmeme_GSTP

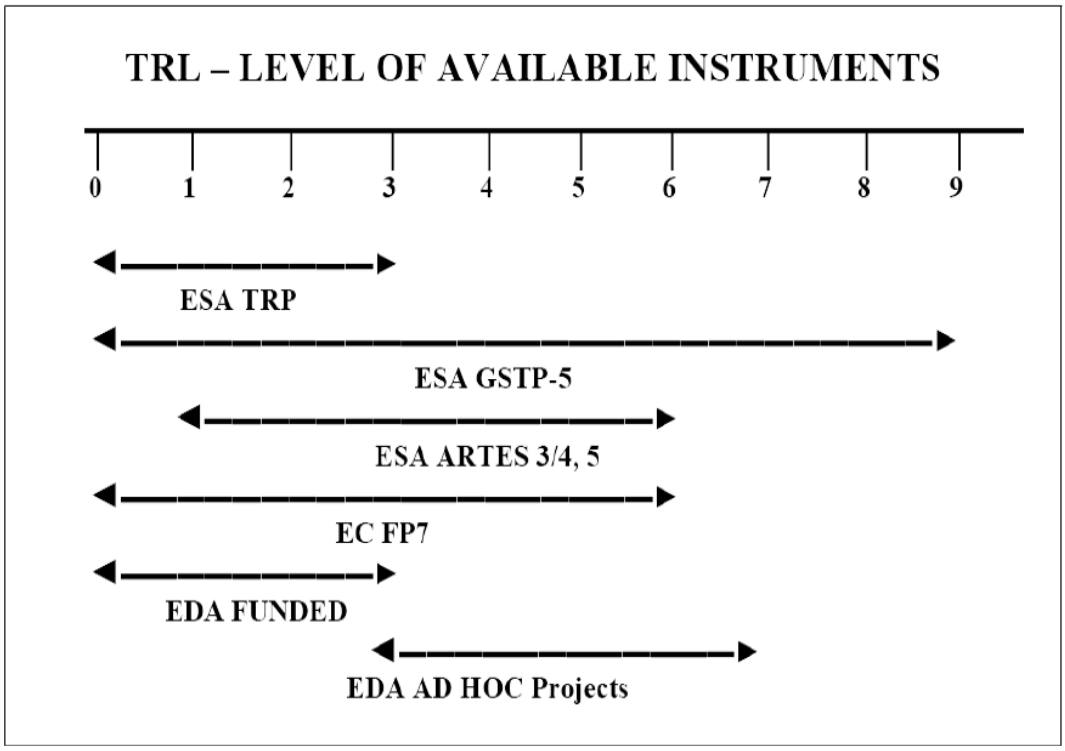


Fig. 4: TRL level within ESA, EDA and EC programmes. Figure from: FP7-Space: R&D activities in support of European microelectronics enabling technologies, presentation by Richard Gilmore, ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011

- *Element 1: General Activities* covers standard GSTP work complementary to TRP and domain-specific technology programmes, maintaining support to industry also with the permanently open Announcement of Opportunities
- *Elem*ent 2: *Building Blocks and Components*, involves the development and qualification to TRL 6 of individual components and related building blocks, in order to be included in the catalogue of European Space Products. This element combines the work of the *European Component Initiative* (ECI) and ESA's participation in the *European Space Component Coordination* (ESCC) for EEE-components development and technological Non-Dependence. (see Chapter 2)
- *Element 3: Security for the Citizens*, covers the development of technologies for systems with security applications.
- *Element 4: In Orbit Demonstration*, makes available in-orbit technology qualification aboard small demonstrator satellites along with the demonstration of new research and operations procedures.

For the GSTP work plan, like that of the TRP, the TECNET working groups prepare a draft with reference to the ESA Long Term plan. It is then approved by delegates of Participating Member States, after prioritising issues that fit with their own industrial plans.[75]

A very important tool in ESA technology management is the *European Space Technology Requirements Database (ESTER)***.** ESTER, also known as *Dossier-0* as it was called on its first issue, is a database that provides a comprehensive view at European level of all envisaged future missions and corresponding top-level technology requirements, including those related to "technology push"[76]. To compile this database, ESA issues a call for future mission and technology wish-lists from all ESA Directorates involved in technology activities, ESA applications programs, National Delegations, major European industries and other interested parties.[77] In this way, ESTER is a key element of the global technology strategy developed by ESA, and a support for the end-to-end technology management processes. The first version of the database was issued in 1999 as a document to be circulated among industry and National Delegations. To further improve the updating

[75] http://www.esa.int/Our_Activities/Technology/About_the_General_Support_Technology_Programmeme_GSTP

[76] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[77] The European Space Technology Requirements Database (ESTER), Dr. Marco Guglielmi, 60th International Astronautical Congress 2009, IAC-09-D3.4.6.

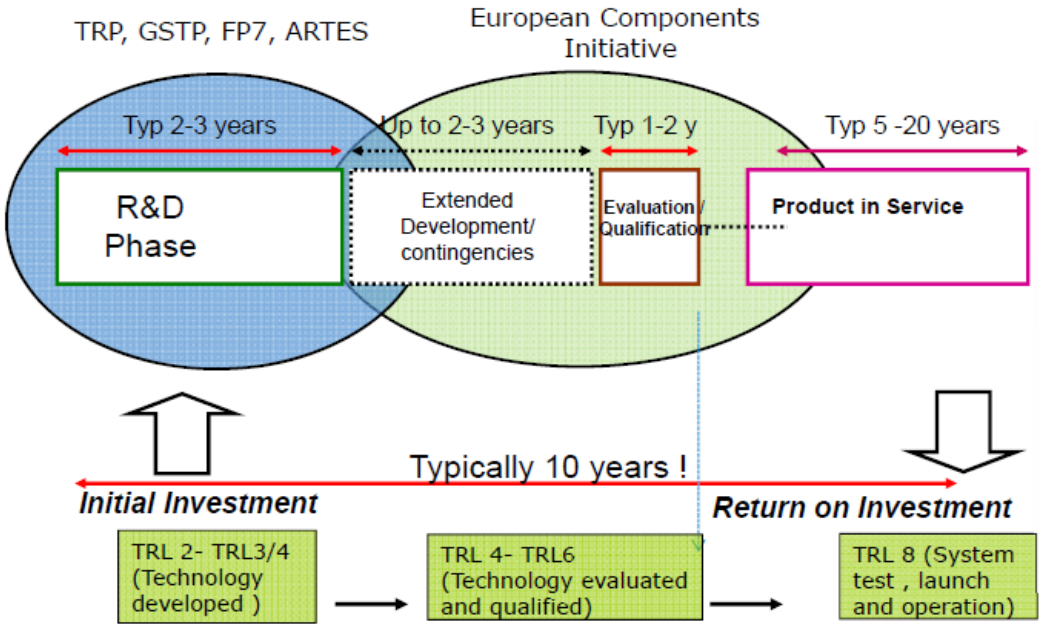


Fig. 5: Timeline from R&D to commercialization. From the presentation *EEE Non-Dependence*, by Mikko Nikkulainen, ESCCON Conference, March 2013, ESA-ESTEC

process and accessibility to ESTER, in 2002 the document was translated into an electronic database available to all registered users in the ESA harmonisation program (see next chapter).[78] Since 2002, ESTER has been updated every 2 years.[79] The database is being considerably expanded and improved over time as processes grow in dimension, participants and complexity.[80]

1.6 Industrial Policy and Competitiveness

On the one hand, technological dependence strongly affects industry at all levels: big companies – prime contractors and subcontractors -, manufacturers and SMEs. On the other hand, a robust industrial base is a prerequisite for Non-Dependence.

Loss of competitiveness undermines both system suppliers and equipment manufacturers. Maintaining current, and developing new, space EEE-component product linesis challenging to justify at the industry corporate level, especially in the current times of economic crisis, where funding is scarce. The reduction of competitiveness is an even more important aspect because of the emerging space faring nations on the global market, with low cost technologies and major technological investment. A crucial element will be the creation of a reinforced and stable institutional market for critical space technologies in Europe. This implies that agencies, and other public and private institutions in charge of financial commitments, support European industry by privileging homemade products. The promotion of such a "buy European" policy would definitely contribute to the availability of technologies and products at the right time and the right price, and at an adequate level of maturity. However, taking into account the small size of the European space market, the possibility of producers and suppliers of components selling outside the EU could be a further incentive for the implementation of more focused strategies both for industry and in the ambit of international relations.[81]

In addition to the points already made the following sets out the further main critical issues for European industry:

Time to Market

Time to market generally involves many years, both for technical reasons and because of obstacles and delay at regulatory level, due to the fragmented nature of the EU. These long times, in a cutting-edge field like space, put Europe into a non-competitive

[78] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[79] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[80] The European Space technology Requirements Database (ESTER), Dr. Marco Guglielmi, 60th International Astronautical Congress 2009, IAC-09-D3.4.6.

[81] European Space Technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

position with respect to its international competitors.

Marketing Strategies for the Individuation of a "Potential Demand" are Needed

The European approach should be translated into a demand- pull mode rather than the current technology – push, in the context of long-term strategies. Industry surveys, product policy and more investments (with effective procurement schemes) are also needed to boost productivity and economic returns. At present, the programmatic tools and institutional framework and coordination among stakeholders are still inadequate. SMEs are one of the foundations of the European industrial system. Often sources of innovation, SMEs could bring high added innovation value to technological development, especially for Non-Dependence objectives, as in most cases SMEs are the suppliers of components and building blocks. Nevertheless, in the aerospace field, SMEs are still not well integrated into the manufacturing process (they are more in the downstream and services sector).

SMEs can play a very relevant role being active in smaller production that would be expensive for larger groups to sustain by themselves. Yet, as a consequence of their reduced size and products volume, continuity of supply and obsolescence can become a problem for SMEs. Special attention must be devoted to supporting SMEs, also at regional level, with contributions harmonised at higher levels by national agencies[82]

1.7 Key Enabling Technologies: New Opportunities for Space

In a cross-cutting and synergistic approach, special consideration should be given to the policies put in place for Key Enabling Technologies (KETs). KETs are "knowledge intensive and associated with high R&D intensity, rapid innovation cycles, high capital expenditure and highly skilled employment. They enable process, goods and service innovation throughout the economy and are of systemic relevance. They are multidisciplinary, cutting across many technology areas with a trend towards convergence and integration. KETs can assist technology leaders in other fields to capitalise on their research effort"[83]. (COM(2012)341)

In 2006, the European Commission identified six technology areas as Key Enabling Technologies:

- Micro-/nanoelectronics,
- Nanotechnology,
- Photonics,
- Advanced materials,
- Industrial biotechnology
- Advanced manufacturing technologies

The first four of the listed KETs are also space enabling technologies.

Although cross-cutting and basic for a number of integrated devices in different areas, KETs activities have suffered from funding cuts due to the economic crisis and the weaknesses inherent in the European industrial system. The last framework programme of research and innovation, *Horizon 2020*, devotes particular attention to KETs. They are one of the major components of the *Leadership in Enabling and Industrial Technology* pillar of the programme. A programmatic approach has been adopted with strategic and operational targets for each of the six KETs. It includes support for both technology-specific and multidisciplinary research and, most importantly, a strongly integrated approach. Also, in the last multi-annual financial framework, the European Commission has proposed a more efficient allocation of EU funding for all the stages of KETs R&D activities. Activities under Horizon 2020, the EU's cohesion policy and policies of the European Investment Bank Group, are therefore being adapted and closely coordinated so that they contribute to the European KETs strategy. In order to coordinate and leverage all these funding instruments, there is a need to combine KETs projects and possible applications fields, primarily through Horizon 2020.[84] It is clear that cooperation between different stakeholders in the industrial value chain could have great potential in terms of partnering and access to markets, bringing effectiveness in technological solutions and reducing costs through co-funded actions. For the purposes of Non-Dependence in space technologies, joint activities with KETs, with the many financial and technological possibilities that have arisen due to new efforts in this field, are strongly encouraged. This was, for example, an added criteria in the second *List of urgent actions for critical technologies* for

[82] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace

[83] *A European strategy for Key Enabling technologies – a bridge to growth and jobs,* communication from the Commission, Brussels, 26.6.2012, COM(2012), 341.

[84] *A European strategy for Key Enabling technologies – a bridge to growth and jobs,* communication from the Commission, Brussels, 26.6.2012, COM(2012), 341.

the years 2012/2013 of the *Joint Task Force for Non-Dependence* (see Chapter 2):

» "An action should address a key enabling technology for Space, addressing a generic, transversal need related to a number of applications"[85]

And also the EUROSPACE Technology Policy Working Group powerfully supports it:

» "The KET could be a tool to fund and support technology developments in key areas such as processors, sensors, optical technologies etc., particularly in those areas where dependence is an issue"[86]

1.8 Defence and Security

As noted above, most space technology is dual-use (*Space qualified assets* are mentioned in the Common Military List of the EU). This intrinsically sensitive nature has meant that, in most space powers, security and defence have been drivers for space, putting it at the highest political level. This is not what has happened in Europe, where space has been affected by a strong deficit of political visibility.

The establishment of a European security and defence policy has taken a long time, with a relatively recent evolution. Europe has only recently achieved a *global* civil and military approach to security and crisis management after a long gestation process. The establishment of the *Common Security and Defence Policy* (CSDP), the founding of the *European Defence Agency* (EDA) in 2004 (with the Lisbon Treaty Art. 42(3) providing the legal basis), and the launch of a *Security Research Programme* under the Seventh Framework Programme of the EU, assigned strategic relevance and full achievement of its dual-use potentialities to space.[87] Another step toward the technological convergence of civil and defence sectors was achieved with the enhancement of the *European Defence and Technological Industrial Base* (EDTIB). Defence-related industries, including space industries, were in this way allowed to restructure and consolidate at transnational level.[88] Although in recent years much progress has been made in this direction, synergies between civil and defence sectors are still not sufficiently exploited.

In this context, the *dependency issue* has acquired increased importance as technological dependence could hinder EU CSDP capabilities. In order to improve the EDTIB, EDA has released contracts for aerospace industries to comprehensively analyse the main dependency issues, with a view to reviewing and adapting the European standards.

Technological Non-Dependence in space is crucial for security and defence. Thus space technology is excluded from the WTO (World Trade Organization) agreement. Space activities and space technology exports are regulated by the governments of space powers and by inter-governmental agreements – for example launcher technology exports are regulated by the MTCR (Missile Technology Control Regime).[89]

The European space industry has a small market share, but with a high export value. European space industry commercial and export sales constitute 45% of its activities[90]. But, as already observed, most space assets are dual-use, and so they are subject to strict defence export control. This implies that a well-defined and balanced regulatory framework on strategic export controls is a precondition for guaranteeing control without hampering competitiveness.

Dual-use goods and technology export controls in Europe are governed by EU Council Regulation 428/2009.[91] Regulation 428/2009 sets common EU rules, a common EU control list and a more harmonised framework, aimed at the free circulation of dual-use items in the EU. In the strategic domain, national rules and restrictions have always been dominant, and therefore fragmented and not harmonised. The objective of this regulation is to overcome this situation by extending the rules of the EU single market also to defence-related items. It is directly applicable in the national legal systems of Member States.

Regulation 428/2009 was accompanied by Directive 2009/43/EC[92] and Directive

[85] *List of urgent actions for critical technologies 2012/2013*, EC-ESA-EDA Joint Task Force
[86] *Space technology in FP8/CSF*, Technology Policy Working Group position paper, May 2011, Eurospace
[87] *The EU Dual Approach To Security and Space – Twenty Years of European Policy Making,* Nunzia Paradiso, ESPI Report 45, August 2013
[88] *The EU Dual Approach To Security and Space – Twenty Years of European Policy Making,* Nunzia Paradiso, ESPI Report 45, August 2013
[89] *Space technology in FP8/CSF*, Eurospace Technology Policy Working Group (TPWG) position paper, May 2011
[90] *State of the European Space industry in 2012,* Eurospace, Space Industry Market Working Group (SIM WG), June 2013
[91] Council Regulation (EC) No 428/2009, Setting up a Community regime for the control of exports, transfer, brokering and transit of dual-use items, 5 May 2009,
[92] Directive 2009/43/EC of the European Parliament and of the Council,
Simplifying terms and conditions of transfers of defense-related products within the Community, 6 May 2009

2009/81/EC[93] (the *defence package*, as it has been called). Directive 2009/43 is on intra-EU transfers of defence products and covers all the defence-related products, components and technologies listed in the Common Military List of the European Union. The aim is to produce simplified rules and procedures applicable to the intra-community transfer of these sensitive items. Directive 2009/81 deals with the coordination of procedures for the award of contracts in the field of defence and security. It introduces new procurement rules in this sector, in order to enhance the internal market.[94] The *defence package* was meant to produce a more harmonised and effective licensing system and to promote cooperation between Member States. Although this is the first simplification of the intra-community transfer of sensitive technologies, European export controls still suffer from fragmentation, and national regulations remain predominant.

Beside the obstacles to overcome inside the boundaries of the Union, the EU export control system meets new challenges with the need to safeguard both security and competitiveness on a global level, taking into account the evolving equilibria for the emergence of new space powers and the rapid growth of technological capabilities. In order to review the export control system, in 2011 the European Commission launched a public consultation for a long term vision for EU strategic export controls with a Green Paper, (COM(2011)393).[95] Member States, the European Parliament, industry associations and economic operators, academia and civil society organisations answered to the call, and a staff working document was released taking into account all the issues that arose from the consultation (SWD(2013) 7).[96]

Export controls impact key sectors of the whole EU economy (not only space and aerospace but also defence and security, navigation and telecommunications, electronics and semiconductors …).

As reported by some industry associations,

> "[…]the aeronautics, space, defence and security industries in Europe represent over 2000 companies with a further 80 000 suppliers, many of which are SMEs, and employ around 696,000 people, with a turnover of EUR 154.7 billion. Much of its activity is affected by strategic export controls".[97]
> Also "[…] in the semi-conductors industry, a significant portion of products and technologies are classified as dual-use. The semi-conductor industry represents around 110,000 jobs directly and up to 500,000 induced jobs in Europe, operating in a worldwide market valued at over USD 298 billion in 2010"[98]

It has been pointed out that, although the most innovative and advanced items are generally produced by a small number of supplying countries and companies, in recent years the availability of many technologies has been growing. This has modified the approach to export controls in Europe, also imposing the participation in multilateral export controls regimes.

The European export control regime is unique, because it combines EU-wide principles with national regulations. Harmonisation is still not satisfactory. In the Member States there are different types of export licensing and different terms and conditions; this is the result of politics and regulatory differences, as well as different levels of technological advancement. This situation brings licensing delays for the EU that undermine the fulfilment of the requirements of the exporting industry and international commitments, hindering market and industrial development.

Another issue that some are concerned about is the much too general character of the definition of criteria for export listing and control, that gives rise to divergences in interpretation.

Some industry associations, companies and academia consider that the EU system is too restrictive, and this strongly affects competitiveness. The main weaknesses underlined are, as mentioned above, licensing delays, and also the lack of transparency of some regulations and the absence of flexibility to be able to react to market and technological evolution.

Another feature considered was the EU control list. In particular, the process for the

[93] Directive 2009/81/EC of the European Parliament and of the Council,
On the coordination of procedures for the award of certain works contracts, supply contracts and service contracts by contracting authorities or entities in the fields of defense and security, and amending Directives 2004/17/EC and 2004/18/EC, 13 July 2009

[94] *Seeking harmonisation: European Space export control at the crossroads*, Rosa Rosanelli, ESPI perspective 54 http://www.espi.or.at/news-archive/757-25-november-2011-espi-perspective-on-european-space-export-control-on-line

[95] Green Paper - *The dual-use export control system of the European Union: ensuring security and competitiveness in a changing world,* 30 June 2011, COM(2011)393

[96] Commission Staff Working Document,
Strategic export controls: ensuring security and competitiveness in a changing world - A report on the public consultation launched under the Green Paper COM(2011) 393, 17 January 2013*,* SWD(2013)7. Hereinafter: SWD(2013)7

[97] SWD(2013)7

[98] SWD(2013)7

update of the EU control list is too long. Moreover, it does not capture the actual technical situation of products, and produces competitive disadvantages.

It has been generally agreed and recognised that to achieve effectiveness in export controls what is needed is "[…] strong and close cooperation between relevant authorities and further coordination and harmonisation to enhance the functioning of the existing system in order to align the rules under which export controls are administered, minimise differences in interpretation and application and ultimately improve response time as well as the substantive assessment of export applications"

The most urgent actions identified through the public consultation process are: the harmonisation of the licensing framework; more transparency and a level-playing field for EU exporters; the enhancement of the exchange of information (although some claimed that a legal basis for this would be excessive and would create only a further administrative burden); enhancement of controls in brokering activities; and the minimisation of additional national controls. Other recommended actions included an updated EU control list, with the exclusion of some non-sensitive and widely available items; the critical re-evaluation of intra-EU transfers; and the revision of the general criteria of criticality in the changing global scenario.

All these issues should be taken into account when considering Non-Dependence. In particular, careful consideration should be given to the issue of how a more equilibrated and effective export control and licensing regime could lead to the intensification of international and transnational cooperation.

In this respect, the staff document stated:

> "Some Member States, industry associations and companies [therefore] argue that strategic export controls should be more conducive to international collaboration in research and innovation and demand specific provisions e.g. to allow for intra-company transfers, to introduce a de-minimis rule that would be appropriate in areas of research & development where typically small samples are shipped for specific testing purposes, and to allow for the supply of items within the framework of international collaboration or large scale projects"[99]

[99] SWD(2013)7

2. The Long Road toward a European Technological Non-Dependence Strategy

This Chapter is devoted to an overview of all the initiatives relevant for Non-Dependence enacted by European actors through the years.

In the first section, the initiatives of ESA, the principal actor for Non-Dependence, are described: the Harmonisation Programme, the European Space Component Coordination (ESCC) and the European Component Initiative (ECI).

In the second section, the initiatives of the European Union are presented: the European Space Technology Platform (ESTP), and some specific actions in the ambit of the Seventh Framework Programme for Research (FP7) and now Horizon2020.

The above mentioned initiatives are single actions at technological level. The further step was the establishment of a Joint Task Force (JTF) for Technological Non-Dependence between EC, ESA and EDA, with these three institutions working for the first time on a joint, programmatic approach for this topic, and putting the Non-Dependence strategy at a higher political level. The last section of this Chapter is devoted to the JTF, the description of the initiating process, the main objectives and the first results achieved.

2.1 Non-Dependence Initiatives within ESA

2.1.1 ESA Technological Harmonisation Programme

Given the many actors involved there is a need to agree on Union-wide objectives and to close gaps and avoid duplication. To achieve these objectives, the *European Space Technology Harmonisation Program* was initiated in 2000, after the Resolution *Shaping the Future of Europe in Space* was adopted at the ESA Ministerial Council in 1999[100]. The following ESA Ministerial Council, in November 2001, reaffirmed ESA's central role in the coordination and harmonisation of European strategy and policy for space technology and invited the Agency and its Member States to pursue the programmatic coordination of technology programs in Europe. The accompanying Ministerial Council Resolution affirmed the perspective in the joint ESA-EC Strategy for Space for strengthening Europe's space potential.[101]

The necessity to identify needs, to map European capabilities and to define common European goals for future development, alongside the identification of opportunities for co-funding, partnership and cooperation schemes, requires the involvement of both institutional and industrial stakeholders. This is why harmonisation is an ESA-led process with the participation, at different levels, of all space players: national agencies and delegations, the European Commission, research institutes, operators, and industry, involving more than 1000 professionals from about 200 companies[102] .

The major objectives of the technology harmonisation process are to[103]:

- fill strategic gaps and reduce unnecessary duplication
- consolidate European strategic capabilities
- achieve a coordinated and committed European Space Technology Policy
- contribute to continuity and coherence between Technology and Industrial Policies

In greater detail, the harmonisation process is focused on:

- identifying technology needs and strategic gaps to fill
- mapping European capabilities and defining common roadmaps for future development
- identifying strategic areas for European independent capabilities

[100] European space technology Master plan 2011, ESA, hereinafter ESTMP 2011

[101] http://esamultimedia.esa.int/docs/harmonisation.pdf

[102] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[103] ESTMP 2011

- creating frameworks of cooperation for coordinated actions
- implementing synergies to avoid duplication and fragmentation

The more general activity is to discuss and define roadmaps and agree on recommendations for future R&D actions and projects, covering all programmatic aspects. Technology Harmonisation gathers and takes into consideration the many different European developments, capabilities and budgets to enhance the interrelated and synergistic roles of the stakeholders in meeting common aims. This process also deals with different situations in terms of technology maturity, industrial competitiveness, funding needs and political interests. Topics under discussion in the harmonisation programme include the identification of sources of finance, their optimisation for space-technology R&D, the management of specialised skills and the possibilities for sharing resources and expertise with potential partners.[104] The contribution of both institutional and industrial players, as well as the understanding of single national issues, improves the effectiveness of technological developments and industrial competitiveness. Moreover, this can guarantee a comprehensive and balanced vision. It is an attempt to overcome fragmentation in an effective way through programmatic action for the establishment of a European technological and industrial policy. This demanding programme naturally leads towards Non-Dependence: harmonisation "*contributes to redressing European dependency through prioritization of strategic and critical technologies developments.*"[105] Assuring European Non-Dependence is one of the goals of the ESA harmonisation process, through agreed implementation schemes for the development of critical technologies and the consolidation of strategic capabilities.

The Technology Harmonisation Advisory Group (THAG)

The *Technology Harmonisation Advisory Group (THAG)* is an ESA delegate body tasked to advise the ESA Industrial Policy Committee (IPC) on technology harmonisation matters.

In 2004 the ESA's Council Working Group on the evolution of the Agency's financial, budgetary and industrial policies (FINPOL) stated the need for a more effective organization in order to promote, in ESA Programmes, the conclusions stemming from the technology harmonisation process for due consideration by Member States and industry. The FINPOL Council Working Group resolution, asking for the adaptation of the technology harmonisation process, was unanimously approved by the ESA Council in June 2004. In May 2005 the IPC approved the Agency Executive's *Proposal for the Adaptation of the Technology Harmonisation Process* and THAG was created with the purpose of implementing the agreed road-mapping process and all the related activities of the programme. It was set up in 2006[106].

The creation of the THAG has been a further step in the formalisation of the ESA harmonisation process. In 2007, the maturity of the programme was outlined in the Communication of the EC *"European Space Policy":* "The ESA-led process of harmonising technology development programmes provides transparency on research across Europe and paves the way for improved coordination" [107]

THAG is composed of 1/2 delegates from each ESA Member State and Canada, but they may be supplemented by experts for specific issues. This body is in charge of the monitoring and implementation of the work plans and roadmaps in the ambit of the program. It also refers to the IPC matters concerning technology strategy and worldwide technology watch, and, at its request, on technology Non-Dependence[108].

After the establishment of the EC-ESA-EDA Joint Task Force for European Technology Non-Dependence in 2008, THAG was expanded to include EDA, EC and their national delegates for specific Non-Dependence issues.[109] This strengthened and formalised the fundamental role of the harmonisation process for the realisation of an European Non-Dependence strategy.

Mapping and Road-Mapping Phase

The Harmonisation process is developed through two cycles per year, each covering half of the topics previously selected. The selection and definition of the technology priorities for a given year engages ESA experts, industry and national delegations. They are in charge of the release of a work-plan defining the number, subject and scope of harmonisation topics based on the following criteria[110]:

- Technology maturity level

[104] http://esamultimedia.esa.int/docs/harmonisation.pdf
[105] ESTMP 2011
[106] http://estp.esa.int/exp/E10549.php
[107] *European Space Policy*, Communication from the Commission to the Council and the European Parliament, COM(2007)212
[108] ESTMP 2011
[109] ESTMP 2011
[110] ESTMP 2011

- Strategic relevance for Europe
- Mission needs and market potential
- Technology gap or unnecessary duplication
- Need to revisit technology

The proposed work-plan passes through the THAG and the IPC, and the final updated version is the basis for the harmonisation process. The roadmaps are at this point prepared with a methodology based on two phases for each selected technology: *mapping* and *road-mapping.*

The *mapping phase* is open to all players. Its objective is to achieve a survey as complete as possible of the technology under examination, taking into consideration the situation inside and outside Europe and identification of the most critical features and suitable partnerships. ESA experts prepare a summary containing[111]:

- A description of the topic state of the art, the exact scope of the harmonisation and the reason for it
- The list of main European players and their products
- The main players outside of Europe and a comparison between European products and non-European ones
- The mission needs, market perspective, technology trends and possibility of spin-in.

This is followed by a meeting open to institutional actors and industry experts, to give their inputs and to produce an updated Technical Dossier as complete as possible. Such a Technical Dossier not only serves as the background for the final convergence to agreed roadmaps, but is also a self- standing document, which gives an account of the situation inside and outside Europe, as a snapshot of the technological landscape.

The preparation of the roadmap is the last step in the harmonisation process. It is based on the *mapping* achievements and data and, unlike the previous phase, the Road-mapping meeting is restricted to the funding parties only (Delegations, EC, ESA). Industrial stakeholders, through *Eurospace*, can in any event produce comments after a first roadmap presentation by the ESA experts appointed for this task and before the final review for approval of the THAG.

The rationale behind the roadmap is to highlight every possible interface and break down the different steps required to attain success into individual 'building blocks', capable of being worked on in a synergistic and collaborative way by the different partners[112]

The final roadmap presents ongoing and future activities, the description of which includes the TRL level and the contribution to Non-Dependence. Moreover, the status, planned budget and schedule, successful components and subsystems that can be reused (to cut costs and increase sustainability), and the best fit of new proposals or implemented programs for future R&D actions are clearly identified[113]. The definitive, updated Roadmap, with the decisions of THAG taken during the Roadmap meeting, is defined and included in the final version of the Technical Dossier. All this is finally endorsed by the ESA IPC and a Conclusive Document is made officially available.

Every year there is an evaluation of the status of all the initiated activities of the roadmap by the *Harmonisation Tracking System (HTS),* constituted by those responsible for the ESA Roadmap and the THAG Delegations for Non-ESA activities.[114] They are tasked to check and report on the level of implementation of the harmonisation process. This encompasses, for each technology, the funding status and the effective actions with respect to the programme activity. Beside HTS, a *Technology Observatory* supports the whole process, providing inputs and feedback.[115]

It should be noted that these roadmaps, obtained through the harmonisation process, are recommendations for decision makers in institutions and agencies, but do not constitute a work-plan to be acted upon. Thanks to their comprehensive perspective, these are fundamental tools for the preparation of ESA and national work-plans, but they are not work-plans as such. As it is, it is not guaranteed that the recommended budgets will be allocated, or that planned actions will be initiated in the programme[116]. The roadmaps only offer a global overview and present the consensus of the entire community of the best route to follow, the opportunities to take and the challenges to tackle.

[111] ESTMP 2011

[112] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[113] ESTMP 2011

[114] ESTMP 2011

[115] European Space Technology Platform – Strategic Research Agenda. Version 1.0 - 22.06.2006 http://www.estp-space.eu

[116] ESTMP 2011

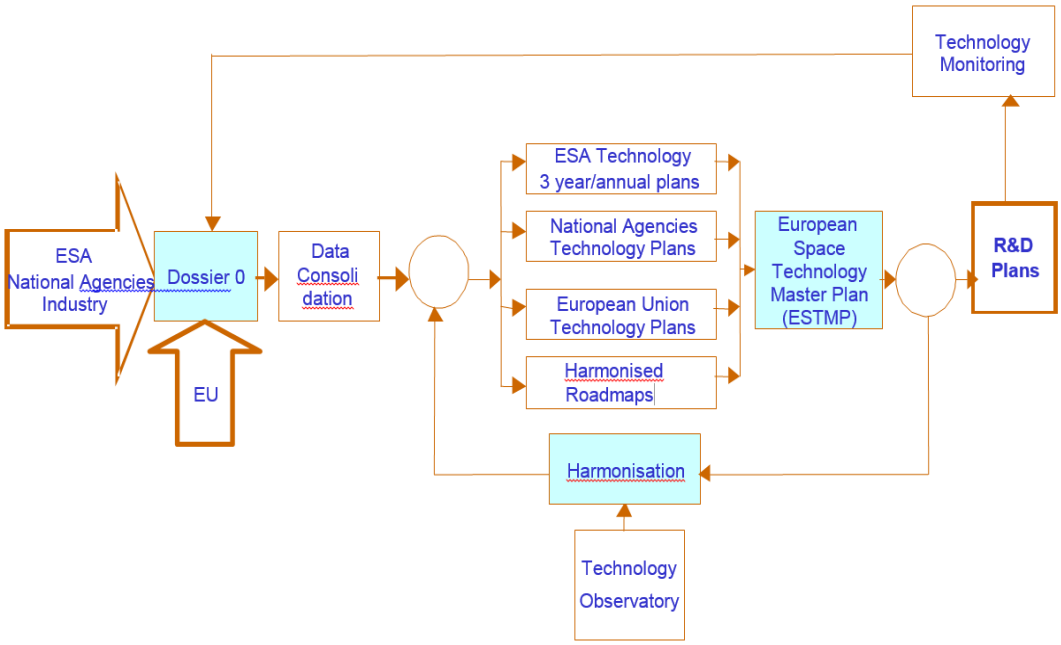


Fig. 6: .The technology harmonisation process.
From European Space Technology Platform strategic Research Agenda, http://www.estp-space

European Space Technology Master Plan - ESTMP

At the end of the mapping and road-mapping procedure, all the data are collected and summarized in the *European Space Technology Master Plan (ESTMP).* The ESTMP was established at the ESA Ministerial Council Meeting in 2001, as a cornerstone and an effective tool for the entire harmonisation process, and its first issue was presented to the IPC in November 2002[117]. It is updated on a yearly basis. Encompassing the main conclusions stemming from the proposed roadmaps together with ESA, national agencies and European Union technology plans, the ESTMP is the most complete and synthesised overview of all planned institutional space-technology activities in Europe combined with industrial requirements. It therefore provides a reference for the next iteration of technology strategies for ESA and its European partners. In R&D terms, the most strategic enabling technologies are prioritized using the ESTMP[118]. In light of this, its establishment can be considered the first broad action made at institutional level toward the establishment of a Non-Dependence strategy.

Upwards of 50 technologies have been harmonised to date and some critical ones involved in the process are among these[119]. It is clear that the harmonisation process paves the way for European technological Non-Dependence and many important steps have been taken in the improvement of the European supply chain and in the management of different resources. Nevertheless, the more general mandate of the harmonisation programme, the huge amount of topics and projects and the inherently strategic nature of Non-Dependence for the EU, have made more targeted actions necessary. This need, also in the light the dynamic global landscape and rapidly growing competition, has long been clear.

In this respect ESA has taken a first step through the establishment of the *European Space Component Coordination* (ESCC) for Electrical, Electronic and Electromechanical components.

2.1.2 European Space Component Coordination - ESCC

European Space Component Coordination – ESCC

Electrical, Electronic and Electro-mechanical (EEE) components have a crucial role in the value chain and life cycles of space systems because of their high enabling use and demanding technological requirements. In 2012, the average European spacecraft had only 35% of European-sourced EEE-

[117] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[118] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation

[119] ESTMP 2011

components.[120] This meant for Europe a situation of very critical dependence (mostly on the U.S.). For these reasons it was immediately clear that, among the priorities for European space activities, EEE-components, their availability and performance advancement, needed special attention and new tools.

A first specific action to achieve a coherent approach in product specification and availability was the *European Space Components Coordination (ESCC)*, a partnership established for cooperation in the field of EEE parts for application in space programmes[121]. In October 2002 the *Founding Act* of the *European Space Components Coordination* was signed between ESA, national space agencies, European space component user industries collectively represented by Eurospace and European EEE space component manufacturers[122].

The ESCC is open to participation by all European space component manufacturers, user industries and European national and international public space organizations. Participating organizations can be Members or Observers. Member Organizations, predominantly Space Agencies together with the European Space Agency, provide an executive function in running the qualification, specification and qualification maintenance activities on a day to day basis.[123] ESA provides the central management function for the Executive, and is both the Qualification Authority and the custodian of the specification system, with procedures, specifications and documents that are made freely available to the international space community[124] (see paragraph *ESCC Executive*). Space organizations that participate as Observers establish a formal link to follow ESCC developments and to access and contribute to some ESCC activities. Observers do not have voting rights in the ESCC, but may indicate their opinion in the decision-making procedures[125]

The objectives of the ESCC, as stated in the Charter of the European Space Component Coordination, are[126]:

- improving the availability of strategically important EEE space components, giving preference to European sources offering competitive performance and costs
- developing and implementing, among users, manufacturers and European national and international public space organizations, a harmonised Research and Development Program for space components, ensuring coherence with other market sectors and cost effectiveness
- establishing an information exchange system with component data provided on a voluntary basis by the Participating Organizations and with access for Industry;
- establishing a European reliability system for space components
- implementing a system for rationalizing the diversity of components for space use, based on the usage of a *European Preferred Parts List*, giving preference to European components offering competitive performance and costs
- enabling the mutual recognition of industrial performance in the various component disciplines, including component engineering, radiation hardness assurance, auditing and inspection, and supporting such recognition by the provision of relevant and regular training opportunities
- promoting component specifications and standards at European level in order to achieve international recognition and to improve industrial efficiency and the competitiveness of the related products on world markets
- maintaining and improve a common and coherent system of component specifications and standards, including related evaluation and qualification programs, techniques and quality provisions, in order to meet user needs and in keeping with market trends

ESCC Structure

The ESCC is organized with a *Harmonisation Task* and an *Executive Task*[127].

The Harmonisation Task is pursued through the following bodies:

[120] EEE Non-Dependence – Mikko Niculainen, presentation to ESCCON conference, March 2013, ESA-ESTEC
[121] https://spacecomponents.org/
[122] http://www.esa.int/Our_Activities/Space_Engineering/European_Space_Components_Coordination
[123] http://www.esa.int/Our_Activities/Space_Engineering/European_Space_Components_Coordination
[124] http://www.esa.int/Our_Activities/Space_Engineering/European_Space_Components_Coordination
[125] *Charter of the European Space Components Coordination*, ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures, hereinafter ESCC 00000, issue 1 October 2002
[126] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[127] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures

- Space Components Steering Board
- ESCC Secretariat
- Component Technology Board
- Policy and Standards Working Group
- Ad hoc Working groups

The Executive Task relies solely on the *ESCC Executive*[128]

The Harmonisation Task

Space Component Steering Board

Overall responsibility for the Harmonisation Task resides with the *Space Components Steering Board* (SCSB), which is the preeminent ESCC body[129]. The SCSB is composed of representatives of the Participating Organizations[130]:

- one representative from each European national and international public space organisation
- four representatives nominated by Eurospace and representing the interests of European EEE space component user industries
- four representatives representing European EEE space component manufacturers
- one representative from each Observer Organization who may attend the meetings of the SCSB

Representatives are entitled to be assisted by a maximum of two advisors.[131]

The SCSB is responsible for the implementation of the objectives of the ESCC. In particular, the SCSB is in charge of:

- defining regularly the overall strategy in the area of European EEE Space components and of issuing recommendations;
- supervising the work of the other bodies of the Harmonisation Task and establishing ad hoc Working Groups for specific tasks with limited duration;
- liaising with the ESCC Executive; organizing at regular intervals the *European Space Components Conference*;
- liaising with other EEE-component organizations; issuing an annual report on the major ESCC activities and achievements and an outline of the major plans for the future[132].

ESCC Secretariat

This is headed by the *Technical Secretary*, designated by the Director General of ESA. The Secretariat gives general secretarial and administrative support to the SCSB, including the provision of permanent liaison and communication links between the SCSB and the other organizational bodies.[133]

Components Technology Board

The Component Technology Board (CTB) is subordinate to the SCSB and charged with the formulation of strategic programs, work plans and multi-year strategic plans for technology research and development in the area of the EEE space components. It harmonises the collectively funded component of research, development, evaluation, qualification, standardization and quality assurance activities. The CTB comprises representatives selected and nominated by the Members to support and contribute to the harmonised and coordinated Technology Research and Development Program.[134]

The CTB provides the basis for the work of SCSB, preparing documents for SCSB study and approval and developing analysis to enable this body to define general strategies and policies. The tasks of the CTB span from evaluating technical issues to assessing market trends, seeking technological breakthroughs and future needs and exploring feasible synergies with other market or technological sectors. In particular, in the implementation of its duties, the CTB identifies critical EEE-components and advanced technologies and suggests means of improving their availability at affordable costs; proposes and supports initiatives for the advancement and promotion of European assembly and packaging capabilities; establishes plans to make component technology available in time for user projects; assesses technologies/components, manufacturer capabilities, technology trends, desired quality level and availability at competitive prices. Moreover, among its responsibilities is the task of identifying the necessary resources, funding, time scales and priorities for the various activities and planning implementation by Members. The CTB also initiates and integrates all enabling activities leading to the formal approval

[128] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[129] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[130] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[131] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[132] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[133] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[134] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures

or qualification process by the ESCC Executive.[135]

Policy and Standards Working Group

The *Policy and Standards Working Group* (PSWG) is subordinate to the SCSB and is charged with the preparation and harmonisation of policy, standards and specification proposals for evaluation, qualification and quality control, leading to a coherent and cost-effective ESCC Specification System for the procurement of EEE space components (see Chapter 3). The PSWG advises the SCSB on any necessary evolution concerning policy, assurance concepts, utilisation and on the potential for harmonisation with other EEE-component systems. The PSWG comprises representatives, selected and nominated by the Members from among experts in the space component field.[136]

The PSWG initiates and supervises studies and research as requested by the SCSB in support of its activities including the provision of component reliability data; monitors application and usage of the ESCC Specification System, as well as elaborates and implements promotional campaigns. Finally, it assists the CTB in the selection of components for evaluation and qualification and supervises the European Space Components Information Exchange System (ESCIES) .and the European Preferred Parts List (EPPL).[137]

The Executive Task

ESCC Executive

The *ESCC Executive* is the entity responsible for the custody and management of the ESCC Specification System and the implementation of its procedures in the evaluation, qualification and certification of components according to the requirements of European component manufacturers.[138]

The duties and functions of the *ESCC Executive* are carried out under the authority of ESA, with the active cooperation and support of other European national and international public space organizations that are Members of ESCC. The Executive is responsible for the harmonisation of the contributions of these organizations in order to ensure uniform implementation of the Executive Task and, in particular, of the ESCC Specification System.

The fulfilment of the ESCC Executive's main tasks and the management and administration of the ESCC Specification System includes many different activities. These comprise: the implementation of the standards, basic and generic specifications, quality controls and rules of procedure approved by the SCSB and adopted by ESA and the European national and international public space organizations; the evaluation and qualification of components, component technologies and component manufacturers, with the further issue of certificates for qualified ones; the performance of audits, reviews, non-conformance controls and other quality assurance and quality control activities; the review and approval of ESCC detailed specifications and provision of support and guidance in the preparation of draft specifications. Finally, the Executive Task provides the European Space community with all information regarding assessed and qualified components, technologies and component manufacturers.[139] The Executive advises the CTB, the PSWG and the SCSB on component policies and programs providing full transparency through progress reports on their status.[140]

European Cooperation for Space Standardization

To achieve the goal of more integrated and higher-quality components and systems, the ESCC works in synergy with the *European Cooperation for Space Standardization (ECSS)*. Established in 1993, ECSS is a cooperative initiative of the European Space Agency, national space agencies and European industry associations for developing and maintaining common standards[141]. The final aim is to provide a single, coherent and recognized set of space standards for use in all European space activities and projects[142]. The existence and availability of such a uniform system of space standards and requirements, accepted and functionally integrated in Europe, is a base for the improvement of technological upgrading and industrial competitiveness and paves the way for international exchanges and collaboration. To date, ECSS activities are recognized and appreciated by all European and non-European space entities and various European and international organizations dealing with standards. Several non-European space agencies

[135] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[136] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures

[138] https://spacecomponents.org/

[139] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[140] ESCC 00000, Issue 1, October 2002, https://spacecomponents.org/specification/docprocedures
[141] *The European Cooperation for Space Standardization*, Luciano Balestra, TRISMAC 2008, http://www.hq.nasa.gov/office/codeq/trismac/apr08/day2/balestra_esa.pdf
[142] http://www.ecss.nl/

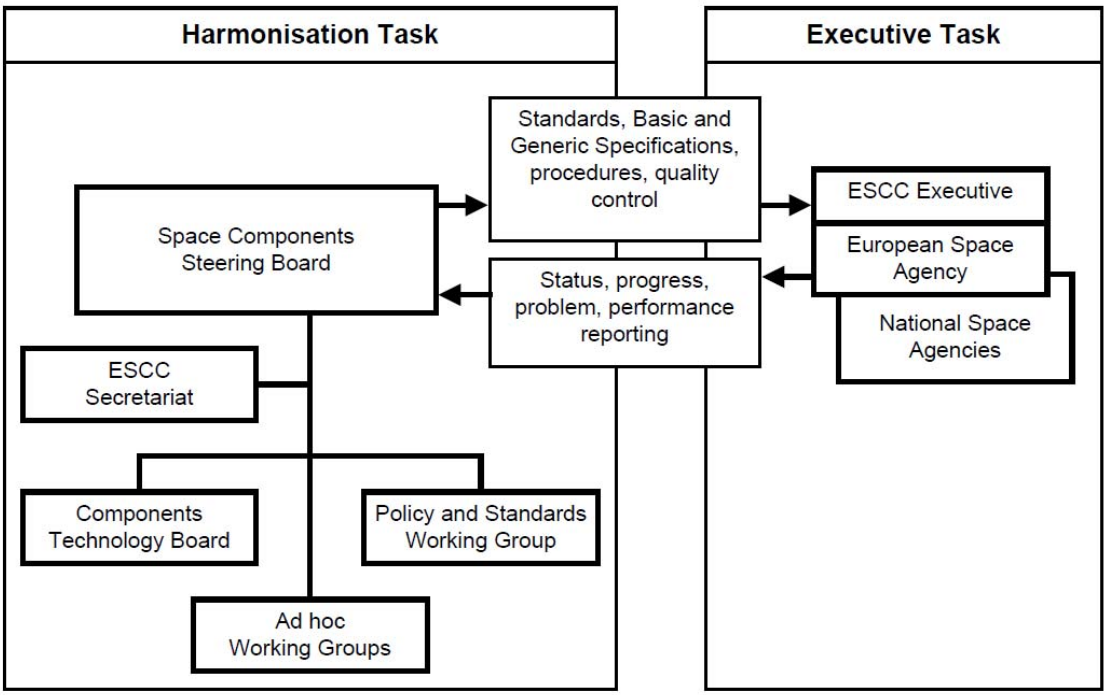


Fig. 7: The ESCC structure: Harmonisation Task and Executive Task. From *Charter of the European Space Components Coordination*, ESCC 00000, Issue 1, October 200

have expressed their interest in using ECSS Documents[143]. Standardisation is certainly a key requirement for the setting up of a Non-Dependence strategy, where international relations should be implemented in parallel with technological enhancement.

ECSS standards are product-oriented[144]. This means that they are defined in terms of what should be carried out rather than in terms of how to organize and perform the necessary work, which is generally the responsibility of the management authorities of single projects and thematic unities and directorates. This allows existing organizational structures and methods to be applied where they are effective, and to evolve as necessary without rewriting the standards. Concerning the component specification system, the ESCC deals with the qualification and procurement of EEE parts, while ECSS addresses the selection and use of the components, in a coordinated and complementary way[145].

The standardization process, after the overall evaluation and qualification of components, is crucially important in the establishment of a technology strategy at European level, for the enhancement of a European supply chain and, of course, for Non-Dependence.

European Preferred Part List (EPPL)

A very important tool within ESCC is the *European Preferred Part List (EPPL)*. It has been established to focus users towards a limited number of component types, covering all design applications for space use, in order to avoid duplication, to achieve cost reduction and to increase procurement effectiveness in a project.[146]

Although the objective of this document is to limit the number of parts, in line with harmonisation and coordination strategies, EPPL is the most extensive component list available, comprising parts from both European and non-European sources, to be further subjected to the qualification process of the ECSS Specification System before being moved to a restricted list of qualified parts.

Making available such an EPPL list actively promotes the selection of parts, for which

[143] *The European Cooperation for Space Standardization*, Luciano Balestra, TRISMAC 2008, http://www.hq.nasa.gov/office/codeq/trismac/apr08/day2/balestra_esa.pdf

[144] *The European Cooperation for Space Standardization*, Luciano Balestra, TRISMAC 2008, http://www.hq.nasa.gov/office/codeq/trismac/apr08/day2/balestra_esa.pdf

[145] *Qualification of passive components according to ESCC*, Denis Lacombe, Lionel Bonora, Olivier Perat, CARTS USA 2011, March 28-31 2011, Jacksonville, FL

[146] The European Preferred Part List (EPPL) and its management, ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures (hereinafter: ESCC12300, Issue 1, December 2006)

there is an established body of data that supports their suitability for space applications, thus enhancing effectiveness and innovative value.

The EPPL is made up of two parts:

Part I: components that are fully qualified or evaluated to recognized space standards and thus have representative data that gives a level of confidence for space usage.

Part II: components that have the potential capability to satisfy space applications requirements, but have not yet reached the level of full confidence and thus require some risk mitigation during procurement.

The inclusion of components in the EPPL is subjected to the fulfilment of a number of requirements. Components allowed:[147]

- are capable of satisfying a wide range of design applications
- are known in terms of technology and show potential for use in flight hardware
- have a significant chance of being utilized for current and future programs
- are available from sources for which there is evidence that they are capable of providing roducts of the required quality
- are freely available on a commercial basis for the ESCC User industry. This precludes the selection of a component subject to any form of selective export control

Priority is given to those parts[148]:

- with an established radiation performance
- available in both through hole and surface mount versions
- suitable for both engineering model and flight hardware
- *available from a European supplier*
- from suppliers whose product lines have been assessed by the ESCC Executive or by another recognized third party authority
- for which a second source is available

For each element type, the relevant manufacturer is listed, together with the applicable procurement specification and relevant characteristics. Moreover, due consideration is given to standardization and, for example, reduction of type variants, values, packages, sizes, application range[149].

A component type must be removed from the EPPL with the emergence of any of the following conditions[150]:

- the component has ceased to meet the general requirements set out above
- the component has become obsolete.
- sources are no longer available or have become subject to export restrictions
- the type has been replaced in the EPPL by a functionally similar but improved component effectively meeting the same characteristics and quality assurance requirements.
- inherent reliability/quality problems have been experienced but not resolved

The EPPL is issued and published in ESCIES, with the approval and under the authority of the SCSB. The SCSB appoints a Technical Authority (TA) to manage the preparation and maintenance of the EPPL. The TA is responsible for ensuring the EPPL is regularly updated with a target frequency of twice per year, while the PSWG is tasked by SCSB with overall supervision of the EPPL activity[151]

The ESCC qualified offer is quite large and covers many current needs. However, interesting activities are ongoing to meet also the future requirements of space applications.

Qualified Part List (QPL), Qualified Manufacturer List (QML)

In the ESCC, user responsibilities are:

- supporting the standardization process embodied within the ESCC Specification System
- initiating and preparing draft ESCC Detail Specifications
- providing feedback to the PSWG on whether the ESCC Specifications are current and adequate and on the need for additions and revisions
- negotiating draft Detailed Specifications with manufacturers[152]

[147] ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures
[148] ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures
[149] ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures
[150] ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures
[151] ESCC12300, Issue 1, December 2006, https://spacecomponents.org/specification/docprocedures
[152] *Procedure for ESCC Document and Specification Management*, ESCC12002, Issue 2, July 2003 https://spacecomponents.org/specification/docprocedures (hereinafter: ESCC12002, Issue 2, July 2003)

Components and parts that are fully qualified are then collected in the *Qualified Parts List (QPL)*. The ESCC Executive publishes a list of components and technologies that have been qualified and capability approved to the requirements of the ESCC Specification System, when the result has been certified by ESA. These components and technologies are selected from the EPPL previously elaborated, and are intended for use in ESA and other European spacecraft and space segment hardware, in accordance with the requirements defined in the ECSS Standard[153].

The ESCC QPL is updated monthly to reflect:[154]

- Additions (new manufacturers, new qualifications, extension of qualified range etc.)
- Deletions (manufacturers, component types, component variants etc.)
- Lapses (qualification certificate validity exceeded)
- Maintenance (extension of qualification certificate validity for a further period)

Components manufacturers participating or supporting ESCC are required to:[155]

- initiate and prepare draft ESCC Detailed Specifications for products they wish to supply to the space market
- agree on the content of draft ESCC Detailed Specifications when the preparer is a component user
- supply products in conformance with published ESCC Specifications
- initiate Document Change Requests, when appropriate, to specifications relating to their products

The Qualified Manufacturer List (QML) operates in the same way. After selection on the basis of the ESCC Specification System Requirements, those manufacturers that have successfully met the ESCC Qualification requirements are subsequently listed in the ESCC *QML*.[156] The ESCC QML is updated whenever significant changes occur to the status of qualified manufacturers, according to the same requirements of addition, deletion, lapses and maintenance of technologies and components produced, validity of certificates and source/producer availability[157]. Both QPL and QML are available for users on the ESCIES website. The QPL and QML are limited to European sources.

2.1.3 European Component Initiative – ECI

Whereas ESCC is a cooperative action intended to create a coherent system of qualification for all EEE-components, and so to reach an overall European technology strategy, the *European Component Initiative (ECI)* was launched, in the ambit of ESCC, with the more specific goal of Non-Dependence in EEE-components and parts types.[158] The objective of ECI is to reduce the dependence of Europe's space sector on non-European EEE-components, in particular those parts provided by single source suppliers or that might became subject to export restrictions.[159] The ECI increases the availability of European EEE-components used in European space missions by identifying critical technologies and then developing capabilities to manufacture them within Europe through the implementation of long-term strategies concentrating on consolidating end user requirements for industry and implementing activities to advance technology levels of components up to space qualified status.[160]

ECI was formally agreed by the ESA Ministerial Council in 2004, with the aim of reversing the trend of dependency. The initiative is an open cooperative project with ESA and several national space agencies each participating with their own funding[161]. ECI was put in place through the collection of *EEE Parts European Action Plans* formulated to develop and qualify components that are strategic for the European space community.[162] All the activities are performed through ESCC. End-user requirements are mapped from the start, thus helping to ensure maximum global market penetration of ECI-developed parts. Members of the European space components industry and equipment manufacturers have the opportunity to propose future technologies for ECI activities through participation to

[153] https://escies.org/webdocument/showArticle?id=119&groupid=6

[154] https://escies.org/webdocument/showArticle?id=119&groupid=6

[155] ESCC12002, Issue 2, July 2003 https://spacecomponents.org/specification/docprocedures

[156] https://escies.org/webdocument/showArticle?id=727&groupid=6

[157] https://escies.org/webdocument/showArticle?id=727&groupid=6

[158] http://www.esa.int/Our_Activities/Technology/European_Component_Initiative_ECI

[159] ESTMP 2011

[160] http://www.esa.int/Our_Activities/Technology/European_Component_Initiative_ECI

[161] ESTMP 2011

[162] ESTMP 2011

the ESCC working groups.[163] Components that have been qualified for space use under ECI funding are formally added to the European Space QPL.[164]

The first goal of ECI is to achieve 50 percent EEE-component procurement from European sources for a typical European spacecraft in the medium term. The subsequent objective is to drastically increase the global market share of European space-qualified component suppliers. With its mission focused on implementation of strategies for EEE, ECI is a further and effective tool for the achievement of a true Non-Dependence innovation policy in the heterogeneous and fragmented European landscape.

Since the beginning of ECI, the number of European qualified manufacturers and qualification certificates has increased notably.[165] To date, four phases of ECI have been approved and initiated[166] (see Annex for details).

2.2 Non-Dependence Initiatives within European Union

2.2.1 European Space Technology Platform - ESTP

The *European Space Technology Platform (ESTP)* is part of the European Commission´s E*uropean Technology Platform Initiative*.[167] Technology Platforms are "industry-led stakeholder fora that develop short-to-long-term research and innovation agendas and roadmaps for action at EU and national level to be supported by both private and public funding."[168] ESTPs were established in 2003 by the European Council in order to strengthen the European Research Area. They seek convergence of technological know-how, industrial capabilities, regulatory and financial institutions so as to finalise a strategic agenda for the development of cutting edge technologies.[169]

In November 2004, the first Space Council[170] was held and the strategic importance of space for the technological development and benefit of European citizens was clearly stated as an institutional goal.[171] With the occurrence of that event, and with reference to the guidelines outlined in the Space Council, the *European Space Technology Platforms* were established after discussions with the Commission's DG-Enterprise Space Policy Unit.[172]

The platform aims to strengthen space technology and its impact in Europe. ESTP is built on the European Space Technology Harmonisation process and shares some of its objectives by complementing and expanding it in many strategic areas[173] It is tasked with creating a direct and efficient connection between technical experts and institutions, in order to gather together the interests and commitments of industrial and institutional stakeholders and increase cooperation in R&D and its effectiveness. In this respect, ESTP aims to foster collaborative research and long-term partnerships for multi-disciplinary cooperation between space and non-space sectors and to promote collaboration in the development of multiple-use technology and applications, with particular regard to the convergence of civil and defence R&D programmes. These achievements are pursued through monitoring activities, ensuring coherence and continuity in investments in space technology responding to the needs and priorities of Europe, coordinating new dynamics, technological initiatives, mapping activities and strategic challenges, and supporting the integration of new EU Member States in the space sector along agreed roadmaps.[174]

ESTP stakeholders include:

- 19 EU Member States plus Switzerland, Norway, and Canada

[163] http://www.esa.int/Our_Activities/Technology/European_Component_Initiative_ECI

[164] http://www.esa.int/Our_Activities/Technology/European_Component_Initiative_ECI

[165] ESTMP 2011

[166] ESTMP 2011

[167] http://estp.esa.int/exp/E10497.php

[168] *Commission Staff Working Document – Strategy for European Technology Platform: ETP2020,* SWD(2013) 272 final (hereinafter; SWD(2013) 272) ftp://ftp.cordis.europa.eu/pub/etp/docs/swd-2013-strategy-etp-2020_en.pdf

[169] SWD(2013) 272 ftp://ftp.cordis.europa.eu/pub/etp/docs/swd-2013-strategy-etp-2020_en.pdf

[170] The Space Council is the meeting of the ESA Council at ministerial level and of the European Union Competitiveness Council (internal Market/Industry/Research)

[171] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

[172] Report of European Research Advisory Board (EURAB) on ETPs.

[173] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

[174] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

- European space industry (over 110 companies)
- Eurospace
- Research laboratories and universities
- The European Space Agency, national space agencies and organisations (ASI, BELSPO, BNSC, CSA, CSO, CDTI, CNES, DLR, DNSC, Enterprise Ireland, FFG, GRICES, HSO, LUXINNOVATION, NIVR, NSC, POLSPACE, ROSA, SNSB, SSO, TEKES)

The objectives of the ESTP, listed in the *Strategic Research Agenda* (SRA), are to:[175]

- Extend the coordination process to all EU Member States
- Provide a long-term vision for space-technology R&D
- Implement at EU level the vision and a coherent cooperation framework for the deployment of space technologies within the European Space Programme
- *Promote actions to reduce European dependence*
- Enhance and cross-fertilise technology (promote multiple-use and spin-in) by:

  Developing synergies between the ESTP and other related non-space Technology Platforms (e.g. in the areas of photovoltaic, fuel-cells, materials, nanotechnologies and nano-systems, telecommunications, etc)

  Promoting joint upstream research on dual-use technologies (civil and security/defence)
- Drive technology R&D strategy in response to EU needs
- Promote the worldwide competitiveness of the European industrial base
- *Facilitate international cooperation on technology issues (also for Non-Dependence).*

The ESTP aims to complement the efforts of the harmonisation programme (and other existing ESA and European programmes) by identifying three strategic pillars:[176]

- *Pillar 1: Non-Dependence* – Development of strategic space technologies needed for Europe's Non-Dependence; promotion of international cooperation for alternative supplies
- *Pillar 2: Multiple-use and Spin-in* – Synergistic actions with the non-space sector in areas of mutual interest (e.g. embedded systems, photovoltaic, fuel cells, nanotechnologies and robotics).
- *Pillar 3: Enabling technologies* – support the implementation of EU policies by developing the technology needed (e.g. in the area of security/defence).

Non-Dependence is the first pillar of the ESTP strategy. With respect to this specific challenge, the Strategic Research Agenda (SRA) recommends to[177]

- Continue the European EEE-component Initiative beyond the current emergency action, based on a sound long-term basis
- Start new initiatives aiming at securing availability of other basic components that might be subject to US export restrictions
- Support the creation of a supplier base for advanced but currently immature technologies (e.g. Compound Semiconductors)

As a conclusive statement about Non-Dependence, the SRA states that:

> "The ultimate answer to this challenge is to guarantee the availability of essential parts and components, ideally required for a range of applications. *Europe will have to provide a capable answer to the European dependency on critical technologies by (1) providing the funding for the necessary in-house development and by (2) fostering international cooperation aiming at diversifying the supply market."*[178]

The ESTP was established to be the prime channel to provide the European Commission with recommendations on a complementary set of actions and funding needed, thus being the more direct link between technical stakeholders, their views and their needs, and European Institutions. One of the main tasks of the ESTP is to seek sufficient funding and to look for more effective implementation of joint activities in the whole Community. Most space technology activities have been developed through the stakeholders' ongoing programmes that are seeking more effective use

[175] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[176] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[177] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[178] European Space Technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu

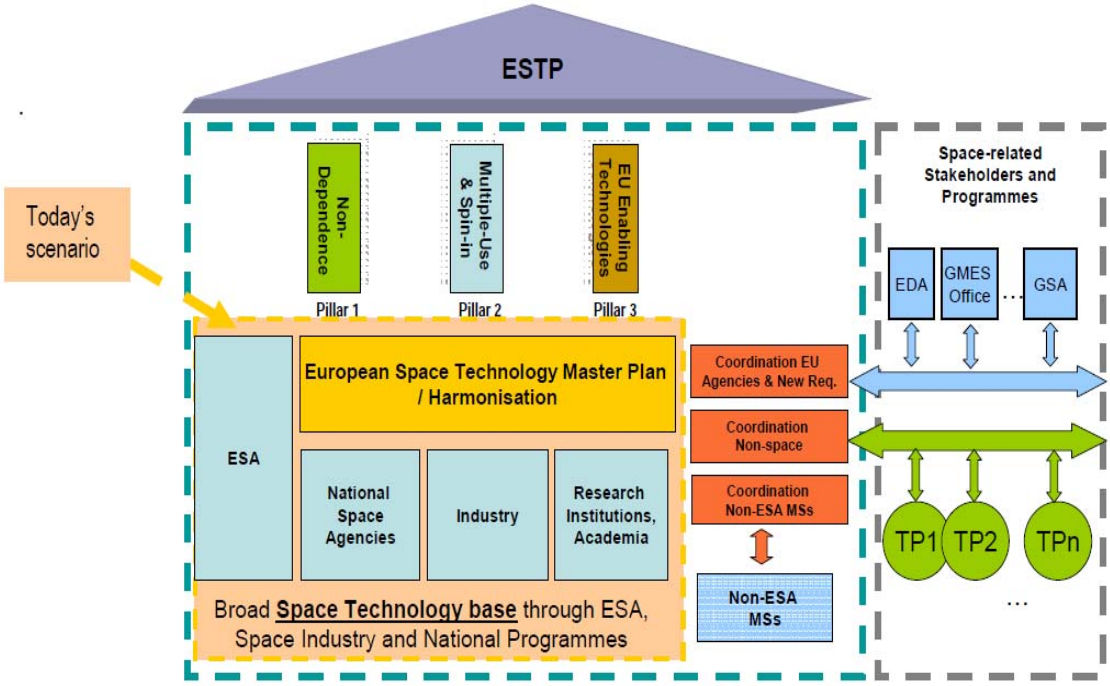


Fig. 8: ESTP in the European technological scenario for Space. Figure from Strategic Research Agenda of *European Space Technology Platform*, Version 1.0, 22.06.2006.

of the different funding mechanisms available, benefiting also from multiple use enabling technologies and spin-in strategies. This synergistic and cross-cutting goal should be strongly developed for the future.

The ESTP was established in phases. The first SRA was produced with strong time constraints to fit into the calendar for the preparation of the *7th Research Framework Programme (FP7, see next paragraph)*. ESTPs have now been recognised by the Commission as a necessary tool to implement Horizon 2020, and to be relevant for the achievement of *Europe2020* and the *Innovation Union* goals.[179] The production of a new *Strategic Research Agenda* for ESTP, updated to take account of the state of the art of technology, the new general context and new strategies and actions also targeted to the new framework program and European initiatives in innovation and competiveness, is strongly needed, in particular by the industrial community.

2.2.2 The 7th Framework Program: First Steps

Initiated in 1984, *Framework Programs (FP)* are the main financial tool of the European Union to support scientific research and technological development activities. FPs are elaborated and proposed by the European Commission and adopted by Council and the European Parliament following a co-decision procedure.

Space was first introduced in the *6th Framework Programme for Research and Technology Development* (FP6, 2002-2006) under the thematic priority: *Aeronautics and Space*.[180]

In accordance with the European Strategy for Space, space-related activities within FP6 have focused on applications and services using space infrastructures, emphasising the possible benefits for society and markets. Support has been given to several areas: Satellite Navigation, Positioning and Timing (in particular in the context of the *Galileo* Programme), Global Monitoring for Environment and Security (GMES, today *Copernicus*), and Satellite Telecommunications.

During FP6, €235 million was available for space over the five years it lasted (with additional funding of space-related activities under the thematic priorities *Information Society Technologies* and *Sustainable Develop-*

[179] SWD(2013) 272 ftp://ftp.cordis.europa.eu/pub/etp/docs/swd-2013-strategy-etp-2020_en.pdf

[180] *FP7-Space: R&D in support of European microelectronics enabling technology*, Richard Gilmore, ESCONN Conference 2011, ESA/ESTEC, 15-17 March 2011

ment, Global Change and Ecosystems, and specific grants for the *Galileo* programme)[181]

With the *7th Framework Program for Research and Technology Development* (FP7, 2007-2013), *space* has had its own thematic priority for the first time. FP7 is also the first framework program over a period of seven instead of five years, in order to be synchronized with the EC's *Multiannual Financial Framework* (MFF).[182] Although FP7 is still mostly focused on infrastructure, applications and services, some efforts and funding have been earmarked for innovation and strategic technology, in a complementary way with ESA and national space programmes (with more coordination and stronger visibility between EC and ESA, addressing gaps and duplications, and giving added value for some target topics). About €1.4 billion has been allocated over seven years, distributed as follows:

- Space-based applications for the benefit of society (mainly GMES) – 85% of budget
- Strengthening *Space Foundations (SSF)* – 15% of budget, all of which dedicated to open calls.[183]

Technology development and scientific research in FP7 are addressed under the Space Foundations title. *Key space technologies, "critical technologies for European Non-Dependence"*, is one of the stated activities of the SSF, and is funded at about 10 million euros per year. It is evident that the budget allocated for every single activity is very limited and not adequate for the effective needs, especially considering the very demanding and strategic value of most of the activities (especially the development of critical technologies).

206 projects have been funded in 5 calls for projects under the 7th Framework Program. Approximately €518 million was spent on Space Research in FP7 between 2007 and 2012 for all the topics in the SSF.[184]

FP7, the first FP since the Lisbon Treaty, has clearly given an impulse to space at the institutional level. However, many weaknesses can be identified in the descriptions of goals and activities, and in particular, there is lack of a true technological plan with consideration of priorities. One of the biggest flaws is that guidelines and directives on space have disregarded the specifics of the space sector business.[185] Moreover, there was little emphasis on the possibility of multiple use as well as spin-in strategies and technology transfer.[186]

Moreover, in FP7 there was fundamental discontinuity in themes from call to call that hindered any strategic action and medium- to long- term strategies, like technological Non-Dependence. In fact, space science and technological projects have to be implemented not only from call to call, but also over successive framework programs, since space roadmaps run over longer timeframes than any institutional program. Such long-term planning must include the allocation of funding, as well as tools for the preparation, maintenance and exploitation of space missions.[187]

FP7 funding was mostly used for collaborative research and coordination between universities with industrial partnerships, rather than for innovation in the industrial sector itself, in this way leaving unresolved the difficulties of the commercialization phase.

Although it is notable that critical space technologies have gained relevance in the goals of the European Framework Programme for Space, the approach has still been inadequate and major attention is needed to develop tailored actions for critical technologies and Non-Dependence, as all experts agreed in the discussion around the further Framework Programme, Horizon 2020.

2.2.3 Horizon 2020: The New Framework Program for Research and Innovation

Horizon 2020 is the latest *Framework Program for Research and Innovation*, for the period 2014-2020. Its definition contains a big novelty: for the first time it combines research and innovation, covering the entire value chain. This is a fundamental change, because it focuses attention on the translation of research and innovation into marketable goods and services, emphasizing the implementation of the "triangle of knowledge" from product concept to market, from academia to industry. As observed many times in the previous chapter, this is the most critical phase and weakness of Europe.

[181] European Space technology Platform – Strategic Research Agenda, Version 1.0, 22.06.2006, http://www.estp-space.eu
[182] *FP7: Space Exploration Research,* Richard Gilmore – COSPAR PEX workshop, 2-4 March 2011, SPI Washington DC
[183] *The FP7 Space Programme 4th Space call*, Massimiliano Esposito, COSMOS Infoday, Padova, 15-16 September, 2010
[184] *Horizon2020/Space*, Héctor Guerrero and Richard Gilmore, ESCCON Conference 2013, ESA/ESTEC, March 14th 2013
[185] *Space technology in FP8/CSF*, Eurospace Technology Policy Working Group (TPWG) position paper
[186] *Space technology in FP8/CSF*, Eurospace Technology Policy Working Group (TPWG) position paper
[187] *Space technology in FP8/CSF*, Eurospace Technology Policy Working Group (TPWG) position paper

In order to fulfill this purpose, Horizon 2020 introduces a major simplification and standardization of funding schemes and strongly promotes cross-cutting and technology transfer activities and the major involvement of SMEs.

The general goal of the Horizon 2020 program is to contribute to the objectives of the *Europe 2020* strategy. It focuses on three pillars: *Excellent Science*, *Industrial Leadership* and *Societal Challenges.* These three priorities are separated but coordinated. Space is included under the Industrial Leadership ambit.

As stated in the Horizon 2020 proposal, "the specific objective of space research and innovation is to foster a competitive and innovative space industry and research community to develop and exploit space infrastructures to meet future Union policy and societal need."

In particular, Horizon 2020 encourages the research community as well as the private sector to intensify their efforts and investments in the space sector; it contributes to supporting a competitive space industry, from manufacture to services, and to providing appropriate infrastructure for the development and exploitation of new technologies.[188] Of the approximately €80 billion allocated for the entire programme, €1.7 billion (about €240 million per year) has been allocated to space.

Non-Dependence is addressed with particular strength in Horizon 2020, thanks also to the establishment of the Joint Task Force of EC-ESA-EDA for European Technological Non-Dependence (see next section) and the draft of the list of urgent actions started in 2009. Clearly, this issue deserves particular attention in this latest framework programme. This was also clearly stated in the recommendations of the FP7 *Space Advisory Group* (SAG), which addressed the specific topic of technology Non-Dependence in several recommendations:

- *Recommendation 3-10:* Horizon 2020 should target the full space industry supply chain in a holistic manner, including the issue of technology Non-Dependence.[189]
- *Recommendation 3-11:* The current process for identification of critical technology Non-Dependence issues should be reviewed. Also, open calls and co-funding are poorly suited to tackling these issues and an urgent remedy should be sought. There should be a policy of European preference for procurement of critical space technologies.[190]
- *Recommendation 3-12:* Horizon 2020 should encourage broader participation from the whole space industry supply chain and should help by reducing bureaucracy and administrative burdens, adopting commercially compatible project timescales, and enabling industry to tackle development risks in a commercially viable manner.[191]

For the period 2014-2015 € 22 million has been already allocated for critical technologies, which will be selected on the basis of the *List of Urgent Actions for Critical Technologies 2012/2013*.[192].

2.3 EC-ESA-EDA Joint Task Force for European Technological Non-Dependence (ETnD)

All the above mentioned initiatives paved the way to a more coherent and less fragmented technological landscape. Nevertheless, what was still missing was a more programmatic approach toward Non-Dependence, in order to focus the attention on the overall scenario. First, the need to join military and civil technology policies for Non-Dependence, with high attention to security aspects. The global dimension of the supply chain was another central topic to be considered, imposing a rethinking and new evaluation of export regulations, especially for dual-use technologies. Moreover, the emergence and development of new space powers modified the old equilibrium, and new risks and possibilities needed to be analysed in the perspective of market dynamics and in that of international relations and foreign policy.

In this complex scenario, single technological advancements, even though significant, are clearly not effective anymore. The new environment calls for a joint, collaborative and long term strategy at the higher political level, setting out a multilateral vision, involving space and non-space, civil and military agencies, institutions and private stakeholders.

[188] http://ec.europa.eu/enterprise/policies/space/research/
[189] *Space research in Horizon 2020 – Recommendations of the FP7 Space Advisory Group*
[190] *Space research in Horizon 2020 – Recommendations of the FP7 Space Advisory Group*
[191] *Space research in Horizon 2020 – Recommendations of t he FP7 Space Advisory Group*
[192] *Space research, Horizon 2020 – work programme preparation*. Presentation by Mats Ljunqvist.

The first step toward such a unified political action was the establishment of the Joint Task Force (JTF) between the European Commission, ESA and the EDA. This joint task force, is an attempt to create a cooperative and unique round table where not only technical, but also political initiatives for Non-Dependence should be discussed with a multi-stakeholder approach. It does not replace any other initiative above mentioned, but it has been established with the intention of giving more emphasis to the political approach of the problem and a tighter and more effective synergy between civil and military institutions, and to produce an operative action by defining a list of few, urgent priorities in the huge technical landscape and fragmented environment. Its activities are based on the work of the many other forums and initiatives already existing for Non-Dependence within and outside ESA.

The establishment of an EC-ESA-EDA Joint Task Force followed a joint workshop between the parties on 9 September 2008 The work of the JTF is based on the ESA Harmonisation program, with the support of Eurospace and SME4SPACE for the industrial community.

Amongst its tasks, the JTF was given 6 months to:[193]

- *define an agreed, common methodology for a coherent Europe-wide approach, building on the existing and recognized processes, such as the ESA led European Space Technology Harmonisation process*
- define a common list of priorities for critical space technologies to be used in 2009, and identify clear instruments for implementation

Considering the time available, an exhaustive list, that would have required a longer timeframe and full consultation with all involved parties, was not drafted. The intention was to work on a first action in view of FP7 -Space Call 3. On 6 March 2009 the JTF agreed on the preliminary list.

The following decisions were taken:[194]

- to launch the European Non-Dependence Process based on the common methodology in the 2nd semester of 2009, with the aim of finalizing a European Non-Dependence list by January 2010 and establishing the process as a structured vehicle for identifying a common European position on the priorities and scope of individual activities to be proposed for funding.
- to review and update the European Non-Dependence list every 2 years
- to monitor the status of the list on a regular basis and for EC, EDA and ESA to meet once a year to review the status of implementation and identify possible improvements in the process.

The work-plan of the JTF uses the same procedure as the Harmonisation program: a first mapping phase, open to all stakeholders, and a second road-mapping phase, only for institutional delegates, finalized to produce the definitive draft of the *List of Urgent Actions*. The first EC-ESA-EDA Non-Dependence Mapping Meeting took place on September 2009, with the participation of member state delegates of the three institutions and industry. A following meeting of the three institutions with industry representatives was held on November 2009 to discuss stakeholder comments on the Draft List. The EC-ESA-EDA Non-Dependence Roadmap Meeting took place in December 2009, with participation restricted to member state delegations and representatives of the three institutions. The list was discussed and the final updated *List of Urgent Actions 2010/2011* was issued at the end of January 2010.[195] The contents of this list was used for the FP7 call for proposals *FP7-SPACE-2011-1 (fourth call).*[196]

Standard Criteria to Evaluate Technologies

In determining the list of urgent actions, technologies are evaluated according to well-defined and agreed requirements fixed by the JTF delegates.

The standard criteria of evaluation are:[197]

- Items shall be of low integration level, i.e. building blocks and components (System/subsystem assembly, methods and tools are not included)
- Items shall have a clearly identified function and performance target

[193] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011, http://www.fp7.org.tr/tubitak_content_files/275/Uzay/Calls/Call4/SpaceCriticalTechnologies_4thCall.PDF

[194] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011, http://www.fp7.org.tr/tubitak_content_files/275/Uzay/Calls/Call4/SpaceCriticalTechnologies_4thCall.PDF

[195] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011.

[196] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011.

[197] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011.

- Items shall be multi-use and/or applications (i.e. not an enabling technology for one shot use)
- Items shall be not available from a consolidated well-assured European source and unrestricted availability from non-European suppliers cannot be assured
- Critical items for which no adequate or sufficient action is ongoing
- As is stated in the Excerpt from Critical Space Technology for European strategic Non-Dependence: List of Urgent Actions for 2010-2011:

» "allocating funds to carry out a particular development activity by itself is not enough. What is, in fact, needed is to have a long term commitment to establishing in Europe the required technical capabilities"[198]

The List of Urgent Actions 2012-2013

The second round of the Joint Task Force process has been completed for the draft of the *European Non-Dependence Urgent Action List* for the year 2012-2013. It started with the mapping meeting in September 2011. A first draft document was released by the three Institutions representatives and delivered for review and comments to member state delegations and industry representatives in November 2011. An updated *Draft European Non-Dependence Urgent Action List*, with industry agreed comments, was then distributed to member state delegations in December, in preparation for the Roadmap Meeting that was held in the same month. After the long work for the implementation of recommendations and comments, the *European Non-Dependence Urgent Action List* was finalised in February 2012 and presented to the respective bodies of EC, ESA and EDA.

In the mapping meeting the importance of focusing the available resources on generic, transversal, multi-use technologies was highlighted. Therefore, some more criteria to add to the standard Non-Dependence requirements applicable for selecting actions on the new list were identified:[199]

- An action should address *a key enabling technology* for space, addressing a generic, transversal need related to a number of applications
- It should be clearly definable and the costs quantifiable
- It needs to differentiate itself from ongoing actions in a clear way and any dependencies on other projects be definable. If applicable, timing in relation to on-going activities should be specified.

The list of selected items was used for *FP7-Space-2013-1 (sixth call).*[200]

What clearly emerged from this entire process, including the description of initiatives and most updated programs at European institutional level, was the need for joint actions, pursuing activities non only at transnational, but also at cross-sectorial level.

[198] Excerpt from Critical Space Technology for European strategic Non-Dependence: list of Urgent Actions for 2010-2011

[199] Excerpt from Critical Space Technologies for European strategic Non-Dependence: list of Urgent Actions 2012-2013.

[200] Excerpt from Critical Space Technologies for European strategic Non-Dependence: list of Urgent Actions 2012-2013.

3. Critical Technologies: State of Play

In this Chapter the state-of-play of the production of critical technologies for Non-Dependence, from a more strictly technical and quantitative point of view, is described.

In the first section, some successful examples of harmonised technologies are given.

Thereafter the macro-areas relevant for future development and investment, with an overall technological view, are summarized.

The final section sets out very specific technologies on which Europe is strongly dependent, with a detailed analysis in this respect.

3.1 Successful Examples of Harmonised Critical Technologies

Upwards of 50 technologies have been harmonised to date including some that are critical[201].

A relevant example is the targeting of a gap in the European market for high-efficiency *Gallium Arsenide multi-junction solar cells* (A GaAs, is a semiconductor compound of the elements gallium and arsenic. It is used, for example, in the manufacture of devices such as microwave frequency integrated circuits, monolithic microwave integrated circuits, infrared light-emitting diodes and solar cells[202]). This technology was primarily procured from the USA.[203] Europe needed to invest in this critical area to gain a competitive position in terms of performance, price and delivery. This was made possible by concentrating resources in a common development programme to achieve large-scale production capability. The initiative was initiated within ESA, with coordination with German and Italian activities.[204] Coordinated research led to the achievement of a European supply chain. European multi-junction cells have since been fitted to the Herschel-Planck space observatory and chosen for Airbus telecommunication satellites. The project, initiated in 2000 with the beginning of the harmonisation programme, was one of the pilot activities[205].

Another notable example relates to *cryogenics technology*, required for cooling ultra-sensitive satellite imagery and with a high critical value. [206] Cryocoolers are used mainly in cold detection chains (detectors, optical components, mirrors) to improve the performance of observation instruments for scientific and Earth Observation satellites, and in telecoms applications. The harmonisation process of this technology has led to European cryocoolers for future Science and Earth Observation missions. Two main industry consortia are now involved in the development of 10-100 K coolers. As the market is not big enough for two players, the aim is to develop the technology either as a specialist industry or as a common industrial approach.[207]

3.2 Current Technological Priorities

In *Agenda 2015 of the Director General of ESA*, *he identifies the top seven key technologies directly* relevant to scientific progress and, in a broader perspective, to the general advancement of Europe in the fast evolving technological landscape. These are: *ultra-stable deployable masts, formation flying and autonomous rendezvous, large monolithic telescopes and mirrors, infrared detectors, lasers, atomic interferometry and optical clocks.*

Beside these items *Agenda 2015* identifies some *key enabling technologies and systems that need* to be developed and allocated to missions in order to be demonstrated. These technologies are expected to bring breakthroughs, also in domains where Europe already shows excellence. Consistent with the goals and priorities identified by ESA, particular attention was given to human spaceflight,

[201] ESTMP 2011
[202] http://en.wikipedia.org/wiki/Gallium_arsenic
[203] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation
[204] http://estp.esa.int/upload/ESA_STP-harmonisation.pdf
[205] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation
[206] http://www.esa.int/Our_Activities/Technology/About_strategy_and_harmonisation
[207] http://estp.esa.int/upload/ESA_STP-harmonisation.pdf

Number of Critical Space Technologies	2006	2008	2010	2012
Total Number of Items (all levels: Levels 1, 2 and 3)	60	108	110	115
Number of Level 3 Items (Levels 3.3, 3.2 and 3.1)	36	69	62	73

Number of Critical Technologies : Criticality Level Breakdown	2006	2008	2010	2012
Level 3.3 (immediate action needed)	28	39	22	27
Level 3.2 (immediate action needed, harmonisation needed first)	1	9	0	5
Level 3.1 (immediate action needed; actions are ongoing)	7	21	40	41
Level 2 (medium term, to be revisited)	18	34	41	39
Level 1 (long term, to be revisited)	6	5	7	3

Table 1: Number of critical space technologies from the year 2006 to the year 2012. From *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee.

exploration and clean space. The named technologies and techniques were:

- *Rendezvous and docking with a non-cooperative target.* Specifically, enhancement of the technologies for the rendezvous and docking of ATV on the ISS
- *Landing techniques* using different technologies depending on the destination, based on current activities (*Entry*, Descendent and Landing *Demonstrator Module* (EDM) of ExoMars, Moon lander) and previous experiences (*Aerodynamic Reentry Demonstrator* (ARD) of Huygens, landing on *Titan,* moon of Saturn)
- *Robotics,* with particular attention to the development of the *European Robotic Arm* ERA,
- *Life-support system*, in particular on the ISS. Through pilot programmes, like *Melissa* for resources recycling, possibilities leading to eco-friendly ground based systems could be explored.
- *New propulsion systems*, to overcome the current obstacles for deep exploration and leading to eco-friendly solutions.

3.3 Status and Evolution of Non-Dependence Situation

The following contains an analysis of the current status of technological Non-Dependence in Europe and its evolution from 2006 to 2012. While the Agenda 2015 listing described above is related to technologies and macro-areas relevant for future development, where it is important to invest, this list refers to very specific technologies in which Europe is strongly dependent, presenting a detailed analysis in this respect. This analysis covers items from all different technology domains and levels of integration. It has been made by ESA building on technology domain surveys.

The main outcomes of the technical analysis are summarised in the following table (the level of criticality for each identified technology item refers to the criticality levels defined previously):

The main outcomes are[208]:

- *115 technology items have been identified as critical technologies,* coming from almost all of the technology domains

The total number of items on levels 1-3 has slightly increased (from 110 to 115) when comparing the 2010 and 2012 values. This stabilisation seems to indicate that the overall number of items has not evolved significantly, and this implies that actions aimed at Non-Dependence are producing the first results

- Among the critical items, *the total number of technologies at the highest level of criticality* (Levels 3.1, 3.2 and 3.3) *has increased from 62 in 2010 to 73 in 2012*
- On the other hand, R&D actions are ongoing for 41 of these items (Level 3.1), which is a very slight increase from the situation in 2010
- There are still 27 Level 3.3 items (needing immediate attention) for which *insufficient actions are ongoing*

As has been seen, the total number of critical technologies is roughly stable, with some cases being addressed through on-going actions and some new cases having arisen. Thus a dynamic situation exists. The emergence of new cases can be related to the many causes listed in Chapter 1, principally changes in the application of export restriction or in the supply chain. The number of items that are considered as critical space

[208] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

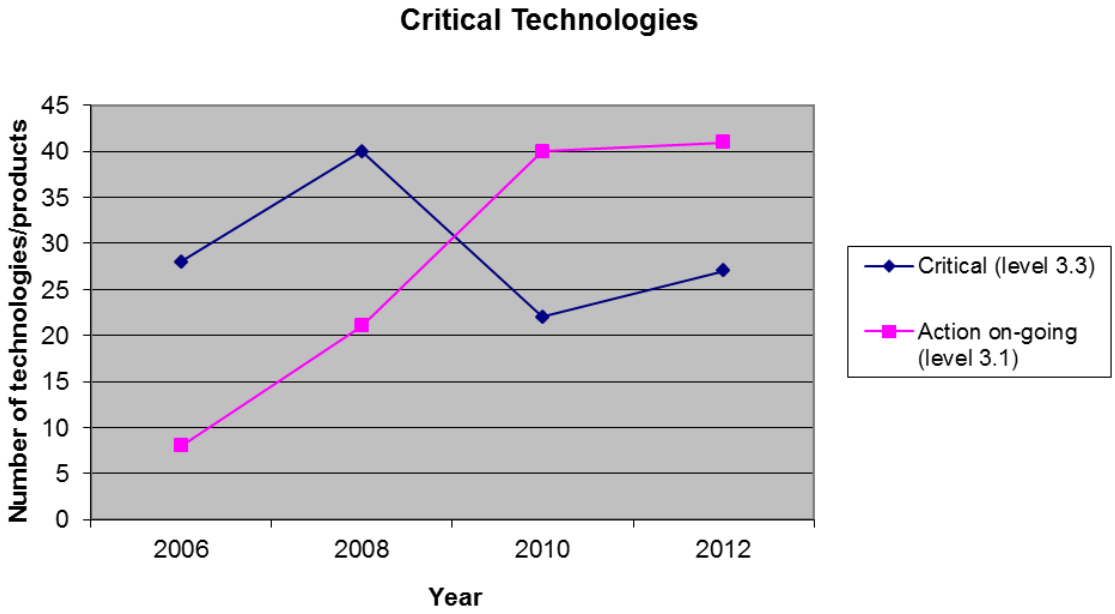


Fig. 9: Evolution of the number of technologies with the highest critical level (Level 3). Figure from *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

technologies in the context of European Non-Dependence is still much too high.[209]

Nonetheless, as stated earlier, the number of critical items for which there are ongoing actions is stabilising, as shown in Fig. 9.

The graph shows that the number of critical items with ongoing actions (through TRP projects, GSTP projects and EU-FP7 projects) has stabilised, rising from 21 in 2008 and 40 in 2010 to 41 in 2012. Also the number of items at the highest criticality level (3.3) has stabilised, moving from 39 in 2008 and 22 in 2010 to 27 in 2012. They comprise 16 technologies for which the situation did not evolve and 7 for which Non-Dependence became more critical between 2010 and 2012.[210]

Although such stabilisations, as well as the new R&D actions taken, are just a first step in the complex process of the establishment of a coherent Non-Dependence strategy for availability and sustainability of technological sources, this is already a good sign.

3.3.1 Technological Domains and the Dependency Issue

The majority of items considered in this Non-Dependence analysis are found in 5 technology domains:[211]

- Propulsion
- On-board Data Systems
- Components - in particular EEE components, but also materials and generic building blocks and parts
- Radio Frequency Payload Systems
- Mechanisms and Tribology

The following chart shows critical technologies for space per discipline. Breakdowns are provided for Level 3 of criticality, and divided for the three different stages: technologies where immediate action is required (Level 3.3, violet), those for which harmonisation is needed (Level 3.2, orange) and those where actions are on-going (Level 3.1, yellow).

[209] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

[210] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

[211] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

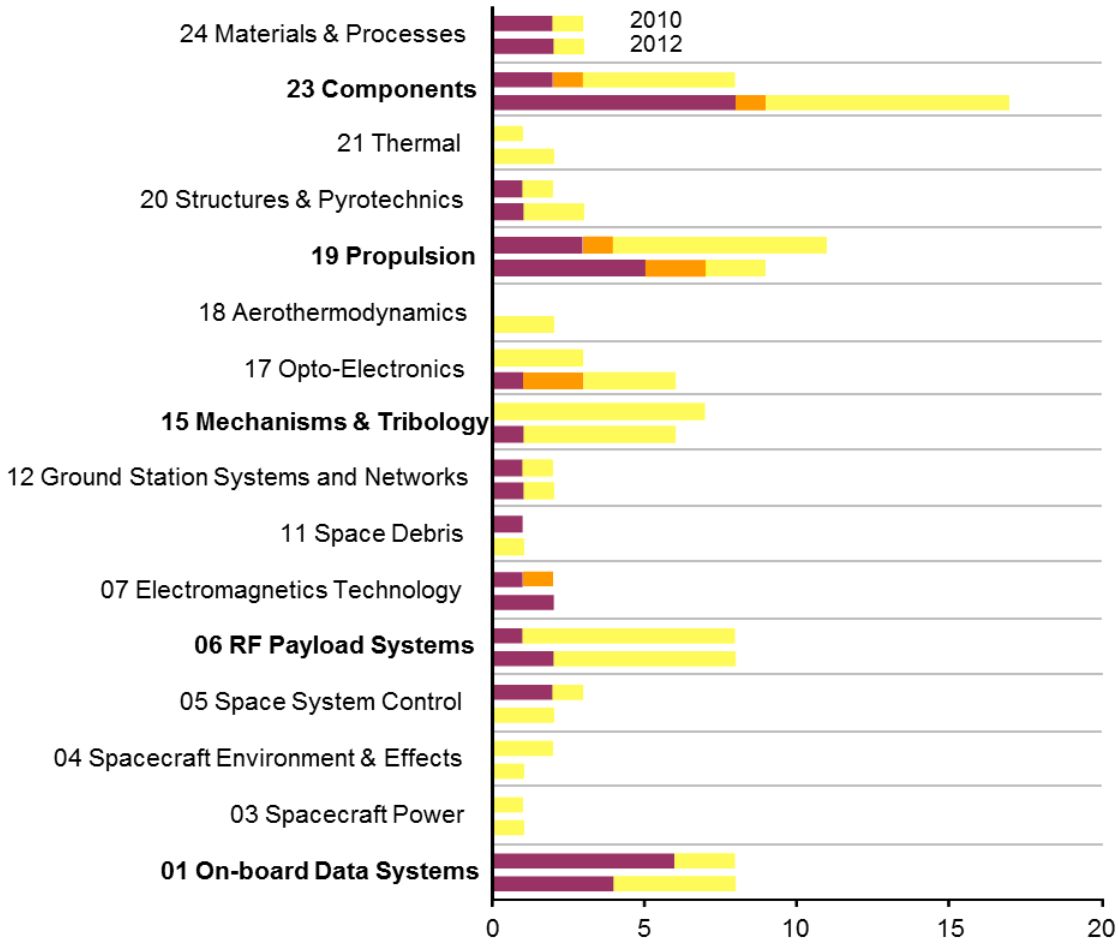


Fig. 10::Breakdown of the number of technology items by technology domain for the highest critical level (Level 3). Figure from *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee.

In the *Propulsion* domain, there are 18 items including 9 items at Level 3 (3.3, 3.2 and 3.1). The main requirements for technological development in this field are mostly focused on higher performance and "green" solutions compliant with the new environmental regulations of the EU. Items identified comprise equipment (thrusters for electrical and cold gas systems and tanks), components (regulators, valves) and also propellants.[212] Among the most relevant needs and technologies to be reinforced, are innovative high performance propulsion systems, hybrid green propulsion, dual mode propulsion systems with Arcjet (REACH compliant) and any substitution of hydrazine[213]

Radio Frequency (RF) payloads and systems are systems using microwave radio frequencies for the space and the ground segments. They comprise not only the specific radio technologies, equipment and systems aboard a spacecraft, but also the ground equipment and telecommunication and transmissions systems through which spacecraft payloads are controlled and telemetry received. They cover advanced functionalities that generally involve digital signal processing and, in some cases, optical technologies. Radio frequency is the preferred technique for the transmission of signals and it is used in a wide range of satellites and applications (scientific, remote sensing, telecommunication, navigation systems).[214] *Tribology* is the science and engineering of interacting surfaces in relative motion; it is a branch of mechanical engineering and materials science.

In the *RF payloads and systems,* and *mechanisms and tribology domains,* there are 14 Level 3 technologies in total, and actions are ongoing for 11 of them. The actions comprise actuators, motors and sensors for mechanisms and specific devices, amplifiers and

[212] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee

[213] Eurospace 2012 R&T priorities

[214] http://www.esa.int/Our_Activities/Space_Engineering/Radio_Frequency_Payload_Systems

dielectric materials for microwave applications.[215]

With respect to *electromagnetic technologies* some criticalities have been identified. The main actions recommended are for the development of radiation-hard, low-noise and large-format detector arrays for operation from X rays to IR and UV spectrometer technology[216]

With respect *to on-board data systems*, there is strong dependency for the key building blocks, in particular EEE. There are 8 Level 3 items including 4 items at the most critical level (Level 3.3). This domain is evolving very quickly, also pushed by non-space markets, but a huge amount of investment is required to escape dependence situations. There has been minimal improvement in this area since 2010. [217]

The last domain is *components*. These items include *parts, materials, generic building blocks and, most of all, EEE-components.* As discussed in previous chapters, this is the domain in which Europe has the biggest weakness and dependence is most critical. *Materials* deserve special attention. From the Non-Dependence perspective, action should be pursued for both availability and performance of components, particularly in relation to lightweight and thermal control. The principal actions in this direction, highlighted in the most updated roadmaps, are in advanced thermal control materials, lightweight structures, materials for large optical structures and mirrors, composite materials (clean space), thermal protection for planetary entry/re-entry challenges and, finally, development of innovative materials[218] Nevertheless, the situation since 2010 appears to be stable.

The *most concerning* change is in the area of EEE-components: there are now 17 Level 3 items, up from 8 in 2010. Of the 17 items, 8 are at the most critical 3.3 level, up from 2 items in 2010. This large increase is partly due to a recent extensive review with the collection of many upgraded items (e.g. solid-state mass memories, components for power) and the introduction of many new items, for example mixed signal components and deep sub-micron processors.

What can be concluded is that, in general, across the different technology domains, the number of Level 3 items has slightly increased, with several of them being new entries. Compared to previous years, the changes are fairly small, with the notable exception of EEE-components.

In order to better understand the situation of EEE-components and the more recent developments, in the following a brief overview of this domain is given of the *European Space Component Coordination* (ESCCON Conference) held in March 2013.

3.4 Main Conclusions of the ESCCON Conference 2013

The last ESCCON Conference, held under the auspices of the ESCC in March 2013, have been identified the main technological criticalities for Europe – and the correspondent state of the art - and their most important features from a strategic perspective. The most relevant required key technologies for the period 2013-2020 were presented, including also ongoing activities and the needs at different levels.

Some important conclusions can be drawn from the overall description:

- Many projects on key EEE-components are ongoing, but funds have been allocated only for a first phase of the production. So there are medium/long term projects, but short term funding. The later phases of the technological process, in particular the standardisation process, the ECSS evaluation, and, in several cases, the harmonisation of a limited number of space qualified critical items, are penalized, and this have a strong impact on the competitiveness.
- Some key EEE-components are at a TRL level that is suitable for fruitful cross-cutting collaboration/joint calls (e.g. Radiation Hard Large *Field Programmable Gate Arrays* (FPGA). This should be strongly supported.

Most of the required key EEE-components are under ITAR restrictions (e.g. MEMS. GaNs). This requires urgent action at technological and political levels, and is a strong incentive for international collaboration. In this context, acooperation between the EU and Japan on the mutual use of EEE qualified parts (which started in 2007) was confirmed and strengthened. Japan will share strategic parts in the EPPL with European customers. In this way, Europe's dependence upon suppliers in the USA has been reduced by 8 %.[219]

[215] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee
[216] Eurospace 2012 R&T priorities
[217] *2012 Analysis of the Non-Dependence situation for critical space technologies in Europe*, Information note, Industrial Policy Committee
[218] Eurospace 2012 R&T priorities
[219] *ECI Update*, Mikko Nikkulainen, 2015

4. Conclusions and Recommendations

- *European Technological Non-Dependence is an overarching topic that must be considered from many different perspectives.* Technological dependence hampers space utilisation, security and defence, productivity and market competitiveness.
- *Even though some technology gaps between Europe, the U.S. and Asia have been reduced, the gap is still much too big for Europe to be independent. This imposes a challenge for foreign policy and advocates international cooperation.*
- The enhancement of the European supply chain through technological development should be pursued, but it should proceed in parallel with international relations and foreign policy actions (e.g. the example of fruitful cooperation with Japan).
- Many initiatives and actions have been taken both within ESA and at institutional level, finally converging in the Joint Task Force of the EC, ESA and EDA for European Technological Non-Dependence. Nevertheless, *an overall programmatic strategy at European level is still lacking.*
- *The first driver for European space technology policy should be security of supply. It should be pursued by two approaches: on the one hand, an enhancement of the European supply chain; on the other hand, the intensification of international collaboration and exchange. The management of the supply chain should be a primary focus in the management of space programmes.*
- The establishment of the Joint Task Force for Non-Dependence has as its goal improvement in the availability of components and building blocks. This is necessary because of the weakness of European industry in this field and the strategic importance of these lower levels of the supply chain. The *Urgent Actions* defined by the Joint Task Force are an important first step.
- *At a technological level, the first priority is to manage medium/long term obsolescence.* It is necessary to achieve the right balance between innovation and product maintenance. To do this, innovation timeframes have to become more compatible with market dynamics.
- *European technological strategies focus principally on product concepts and the early stages of innovation. More attention must be given to maturity and readiness aspects (high TRL).* This is the guarantee that the European space industrial base is able to supply the technologies required, and thus a crucial step in remaining competitive in the global market. In this regard, qualification and in-orbit validation are fundamental steps. Strong efforts and investment should be applied to these later phases of the technological process, to bring innovative technologies to the right level for risk-free implementation in European programmes and for commercialisation. Intensifying the frequency of control and qualification activities will bring more efficiency in the technological process and in the planning capability.
- The periodic preparation of detailed technical roadmaps, marking the evolution of activities and perspectives, is a very important tool, especially when considering the fragmented European reality and the large number of stakeholders involved. Similarly, it is essential to standardise technologies and procedures. The definition of standards and of precise final requirements is a precondition for transnational and international collaboration, both within EU and with third countries.
- A high level of reliability is one of the main characteristics of European technologies and missions, and actually contributes to giving Europe an outstanding position in the global space arena. This is an achievement that should be maintained and improved, although it requires long timeframes and expensive technical processes.
- *The current European fragmentation, at national and industrial level, strongly affects the technological scenario. More coordination is needed,* with tailored actions, joint undertakings, strong synergistic efforts and technology transfer plans. Technology planning and coordination is fundamental, as is the elaboration of long-term prospects for breakthrough concepts and associated technologies. The involvement of all the different actors in a structured multi-stakeholder approach,

especially in the current situation, is crucial for achieving simplification, and through this effectiveness and progress.

- Europe should invest in its human capital to ensure not only availability, but also (and mostly) capability. This is the key to maintaining a central position in the global context, also in the light of the huge investments in human capital of the emerging Asian space powers.
- *To boost the development of the space sector, stakeholders must make use of all EU, national and regional initiatives that promote synergy in space activities. Since space assets are integrated systems of many different components, a cross-sectorial approach is essential.* Spin-in activities should be promoted and enhanced, in a fully open innovation approach. This is relevant to increase the achievement of Non-Dependence goals. Within the ambit of ESA there are many programmes devoted to such cross-cutting promotion. It is necessary to continue to invest in this direction. A well-defined planning of such synergistic actions should be pursued. As concerns components, nanotechnologies and advanced materials, space activities can leverage the many actions proposed by the EC on *Key Enabling Technologies* KETs (for example with joint calls). In this respect, the role and responsibilities of the ESTP are still insufficiently exploited.
- Public support for R&D in space technologies and long-term space programmes is essential because of the high costs, high level of risk and the insufficient returns from commercial markets. All the different funding sources available should be streamlined and harmonised to achieve more appropriate and effective mechanisms, tailored for longer timeframes and specific programmes.
- *Regional and national funding should be harmonised with the technology plans of national space agencies.*
- *In addition to security of supply, and thus the availability of technology, the sustainability of product lines must be ensured. A sustainable market is a fundamental condition for being competitive*. Domestic European markets are limited and fragmented. They need to be aggregated. This should include not only the national space markets, but also be combined with other collateral sectors, first of all defence. Synergies between the civil and defence community are still not being sufficiently exploited. They should be promoted and improved to attain a fully effective Non-Dependence strategy.
- *European industrial policy strongly affects technological Non-Dependence. The inefficient translation of the knowledge base into products and services, the excessively long time to market, as well as the fragmentation of industrial and regulatory national systems are among the most important factors that directly lead to obsolescence and dependency issues.* The combined product policy to link technology evolution to user requirements and to procurement programmes must be enhanced.
- *Horizon 2020*, the framework programme for research and innovation, aims to promote the *triangle of knowledge* by also looking at the commercialisation phase, including through better involvement of SMEs. In the establishment of a Non-Dependence strategy at European level, all the new possibilities provided by the new programme should be fully utilised.
- Although the most innovative and advanced items are generally produced by a small number of supplying countries and companies, *in recent years the availability of many technologies has been growing. This has modified the approach to export controls in Europe, also imposing the participation in multilateral export controls regimes.*
- *The European export control regime is unique, because it combines EU-wide principles with national regulations. However, harmonisation is still not satisfactory.* In the Member States there are different types of export licensing and different terms and conditions; this is the result of policy variations and regulatory differences (beside different technological advancement). This situation brings licensing delays for the EU that undermine the fulfilment of the requirements of the export industry and international commitments, hindering market and industrial development. Harmonisation of national legal frameworks should be combined with flexibility and consistency at European level, in order to achieve the best balance between security and productivity.
- The establishment of a well-balanced and harmonised regulatory framework is at the basis of the evolution of any Non-Dependence strategy. This refers firstly to export controls, but procurement regulations and intellectual property rights are also crucial in this specific context.

Annex

A.1 European Component Initiative (ECI): The Four Phases

ECI PHASE 1: 2004-2012

This first phase aimed to reduce *dependence on the supply of EEE-components from sources subject to export restrictions,* seeking compatible replacements for US-ITAR devices.[220] This first phase was financed by ESA, the French Space Agency CNES and Germany's DLR, the two national space agencies that immediately joined the ECI initiative. In particular, about 20 different components were developed thanks to ESA-funding, and another 11 were sponsored by CNES and DLR. The majority of these were included on the Qualified Parts List of the ESCC, endorsing them for use on European Space missions.[221]

Key developments: Power Mosfets, Fuses, Relays, MMICs, Mixers, PLL[222]

ECI Phase 2: 2009 – 2011

This phase had the aim of *providing competitive alternatives, in terms of costs and time to market, in Europe.* The work-plan addressed two product categories[223]:

- *Product technologies* – where the deliverable is itself a space qualified component replacing the non-European product
- *Enabling capabilities* – where the deliverable technology enables processes and capabilities to produce space qualified components

The yearly budget allocated for phase 2 was 6.5 million euros.[224]

Starting in 2006, the European space components industry and equipment manufacturers, through their membership of the ESCC Components Technology Board and some ad hoc working groups, collected a list of possible part types for the ECI Phase 2 considering the new objective and guidelines. This list identified 54 new and replacement part types.[225] As stated above, *the general criterion for the items selected was to ensure deliverables competitive in terms of pricing, performance, timing and maximum global market penetration for the ECI developed parts,* as these are some of the key strategic elements for *EEE-components' Non-Dependence.* In addition to the active dialogue between ESA and component and equipment manufacturers under the ESCC umbrella, a specific working group under the CTB carefully analysed each of the part types on the list with respect to a set of measurable parameters, in order to provide an overall priority for each device. These criteria took into account the market potential, ITAR or other restrictions, technology maturity, predicted life for the product line and the development investment against the potential returns.[226] The highest priority items were identified and many of the ECI activities are included within ESA's General Support Technology Programme (TRL 8-9, see Chapter 1).

Key developments: MMICs, PLL(s), Capacitors, Fuses, Optical connectors. FPGA(s)[227]

ECI PHASE 3: 2011 - 2013

This phase had the aim of providing longer term access to strategic components and

[220] ESTMP 2011

[221] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[222] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[223] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[224] ESTMP 2011

[225] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[226] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[227] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

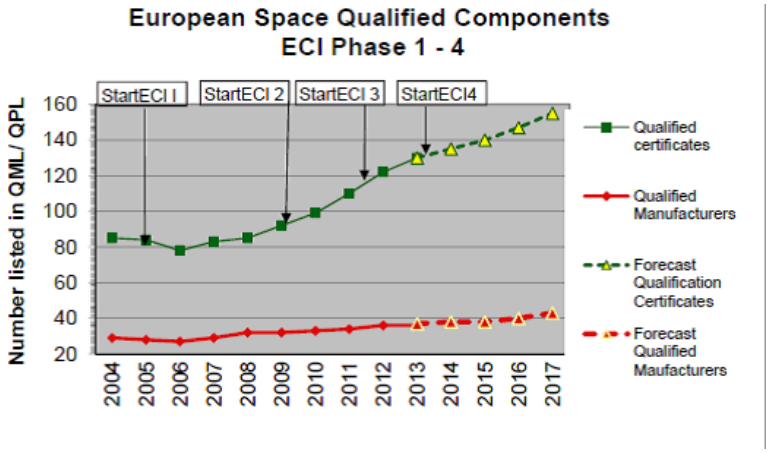


Fig. 11: ECI four phases. From: *EEE Non-Dependence* – Mikko Niculainen, presentation to ESCCON conference, March 2013, ESA-ESTEC

technologies.[228] Since only a limited number of the selected activities could be started under ECI phase 2, and the need to review the requirements of business and technology was still urgent, in 2010 a complete survey of future needs was produced, as usual in the context of the ESCC. The result was a new set of priority items, and the need to consider key strategic technologies for the future was stressed. As in phase 2, ECI phase 3 included, enabling technologies and conventional replacement parts with this highly critical characteristic.[229] The yearly budget for phase 3 was 13.5 million euros

Key developments: DSM, large FPGA, High Pin Count assembly technologies[230]

ECI PHASE 4: 2013 – 2016

With the description "*Key strategic components for Space in Europe*", ECI phase 4 builds on phase 3, with the aim of l*ooking further ahead at the technologies that will be needed in Europe by 2020, in order to strengthen the European supply chain.*[231] This will enable Europe to meet the necessary advances in technology while becoming less dependent and more competitive in the global market.[232]

The required European components and technologies are well defined and have been agreed by ESA, national space agencies, component suppliers, equipment providers and end users for the next 5-7 years.[233] Nevertheless, the ECI is funded only up to the end of 2015 and at a lower level than required to maintain and develop technical capabilities. The shortfall to complete the existing technology roadmaps is approximately 10 million euros per annum.[234]

The required key technologies in the period 2013-2020 are listed below[235].

- Space qualified supply chain for 65nm Deep SubMicron ASIC technology
- ADC/DAC, Flip-chips, high pin count packaging
- Space qualified Rad hard Large reprogrammable European FPGA
- European space qualified Gallium Nitride (GaN) supply chain
- Space qualification of European Mixed Signal ASIC technology
- Extended range of advanced VLSI products
- DC-DC Converters, MOSFETS, Pulse Width Modulators, Line drivers etc…
- Next Generation general purpose Micro-Processor (NGMP)
- Next Generation Digital Signal Processor (DSP)
- European high performance CMOS image sensor technology supply chain
- Space qualified European RF MEMS process

[228] http://www.esa.int/Our_Activities/Technology/European_Component_Initiative_ECI

[229] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[230] https://spacecomponents.org/webdocument/showArticle?id=899&groupid=7

[231] ESTMP 2011

[232] ESTMP 2011

[233] EEE Non-Dependence – Mikko Niculainen, presentation to ESCCON conference, March 2013, ESA-ESTEC

[234] EEE Non-Dependence – Mikko Niculainen, presentation to ESCCON conference, March 2013, ESA-ESTEC

[235] EEE Non-Dependence – Mikko Niculainen, presentation to ESCCON conference, March 2013, ESA-ESTEC

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